QUANTUM CONFINEMENT OF NANOSTRUCTURED SYSTEMS

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"I would rather discover a single fact, even a small one, than debate the great issues at length without discovering anything at all."

- Galileo Galilei

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ABSTRACT

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by Buddhi M. Rai

The evolution of modern electronic devices relies upon semiconductor heterostructures. More recent progress in material growth has made it possible to create a myriad of heterostructures by combining not only different semiconductors but also semiconductor and complex oxides. This has opened up new potential for optoelectronics, quantum information and spintronics.

The layer thicknesses and chemical composition in heterostructures can be controlled over a length scale comparable to or smaller than the electron de Broglie wavelength. This results in novel quantum effects related to size. Quantum well structures can be designed to perform special functions that are the core of many modern quantum devices.

This project has been conducted to explore the electronic properties of nanostructured materials that are artificially created and mimic prototypical quantum mechanics problems. The main focus was modeling quantum confinement in a variety of 1-D potential profiles using a simple, physically transparent tight binding scheme.

We analyzed the electronic structure of quantum well (QW), double QW and superlattices. We also investigated structures that reproduce the harmonic oscillator, the Mathieu and triangular potential wells as well as the potential profile of a hypothetical quantum cascade device. The numerical results are compared with analytical solutions when available. Charge rearrangement has been investigated in the case of δ-doping and doped QW.

We demonstrated that tight binding modeling is qualitatively able to capture the electronic properties of nanostructures and provide a good pedagogical way to introduce novel concepts in quantum mechanics and solid state physics.

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CHAPTER I

INTRODUCTION

1.1 Modern perspective of semiconductor physics

The band-structure engineering of artificially created layered materials (heterostructures) has led to a new generation of semiconductor devices. They are based on quantum well structures and their operation and performance is controlled by quantum effects. For example, quantum well laser and high electron mobility transistor (HEMT) are well-developed quantum devices. Quantum well lasers are extensively employed in optical communication systems, bar-code scanners, laser arrays and optical disc readers/writers; whereas HEMTs are used in low-noise receivers, digital integrated circuits and microwave circuits [1].

The development of epitaxial growth techniques, such as Molecular Beam Epitaxy (MBE) and Metal-Organic Vapor-Phase Epitaxy (MOVPE), has made it possible to control the chemical composition and crystal structure at the nanoscale level for a variety of diverse materials including magnetic multilayers and metal-semiconductor nanostructures [2, 3]. The epitaxial techniques modify the electronic properties of the constituent bulk layer-materials by creating novel nanoscale systems [4].

The electronic structure of the quantum well (QW) is one of the basic problems of quantum mechanics, and it is described by the wave functions Ψ_i and the energies E_i as derived from the Schroedinger equation, $H\Psi_i = E_i\Psi_i$, where H is the Hamiltonian of the system. The problem is usually used to introduce quantum effects such as quantum confinement and quantum tunneling.

Electrons possess not only charge but also spin. The interplay of such properties of an electron incorporated with the confinement effect is interesting scientifically and technologically. The exploitation of the spin properties of an electron has motivated research in spintronics [5]. The discovery of high spin source materials such as half-metals in the near past has excited considerable interest in combining such systems with semiconductors [3, 6, 7]. Therefore, the investigating the electronic structure of the QW has tremendous relevance as a model of modern nanostructures for electronic, optical and spintronic applications. The main problem is trapping an electron and dictating it to the intelligent action as we desire.

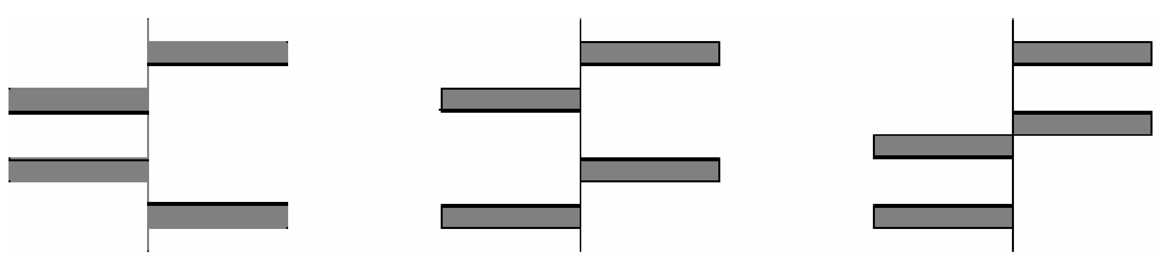
1.2 Scope and Objectives

The goal of this research is to investigate the basic properties of the quantum well, for example the quantum confinement effect, in nanostructured systems. The energy levels and electron wavefunctions of quantum wells are evaluated by considering several interesting potential profiles. The problem will be undertaken using a simple **tight binding approach** which allows one to explore the electronic properties from a model approach [8]. Artificial materials created by combining semiconductors like Si, Ge, GaAs, AlAs are critically important not only because of the growing interest in nano-science research but also their technological role in information storage and processing. The goal of this study is to find ways to estimate the properties of simple quantum systems. The research will provide simple ways to describe the basic principles and the relevant practical measures which can guide the theoretical developments of electronic structure of the quantum well. An additional goal is to provide a practical approach to quantum mechanics and solid state physics.

1.3 Heterojunction - a building block of heterostructures

The interface arising by joining two different semi-infinite semiconductors is called a heterojunction [9]. The forbidden energy gaps of the constituent layers of the semiconductors are usually different, and the materials' electronic properties abruptly change at the interface plane. It is desirable for the difference of lattice constant between the constituent semiconductors to be about (0.01 $\overset{0}{A}$) 1 per cent or less of the lattice parameter. Fortunately, many semiconductor materials crystallize in the diamond or zinc-blende symmetry and satisfy the lattice matching requirement [10]. The electronic properties depend largely on the band offsets and on the bulk forbidden gaps of the constituents [1].

The band alignment determines the properties of heterojunctions. The topology of the band alignments is classified according to the relative ordering of the band-edge energies as shown in the Figure 1.1.



(a) (b) (c)

Figure 1.1 Diversity of band alignment (a) straddling (b) staggered (c) broken gap

The pattern of alignment in Figure 1.1(a) is called *straddling* configuration. The remaining types, given in Figures 1.1(b) and (c), are known as *staggered* and *broken gap.* From the superlattice engineering point of view, straddling is known as Type-I and staggered and broken gap as Type-II.

1.4 Heterostructures – quantum well

Heterostructures are formed from multiple heterojunctions. The double junction structures are known as quantum wells (QW) and the multi-junction structures as superlattices (SL) or multiple quantum wells (MQW). In Type-I quantum wells electrons and holes are confined in the same layers; however, in Type-II quantum wells electrons and holes are not spatially confined in the same region. The recombination times of electrons and holes are long in Type-II heterostructures. The electron (solid circle) and hole (open circle) confinement in Type-I and Type-II single quantum wells are shown in Figure 1.2 [11].

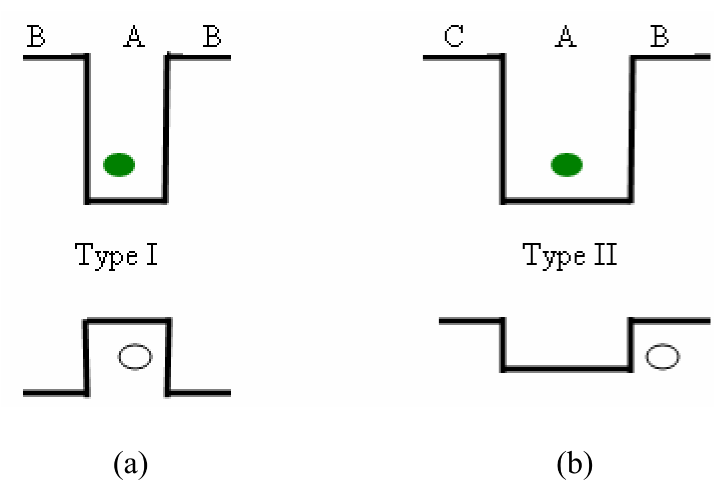


Figure 1.2 Single quantum well (a) Type-I (b) Type-II

1.5 Superlattice and multiple quantum well

A superlattice is a set of quantum wells which are sufficiently closely spaced for the wavefunction, representing the electrons, to tunnel between the wells. The model of the superlattices' electronic structure is based on the assumption that the separation between adjacent wells is small enough for the wave function overlap to be finite. However, the overlap may be regarded as a small correction if the wave function is highly localized in the well. A microstructure in such situation is called a multiple quantum well (MQW). The wavefunction becomes very small between the wells in MQWs [9]. When the alloy layers (barriers of SL) are sufficiently thick (>100 $\overset{0}{A}$), the tails of the wave functions associated with the states confined in the individual wells do not penetrate far enough into the barrier for them to overlap; as a result, the system will not be able to form a superlattice, but a MQW. The potential profile of the superlattices is analogous to the Kronig-Penney model [12].

QWs have applications in lasers, LEDs, etc; while superlattices have a variety of applications in resonant tunneling structures and far-infrared detectors [10, 13]; likewise, MQW structures have applications in various electronic devices, including quantum-cascade lasers and solar cells [14, 15].

1.6 Conception

Basic quantum mechanical problems such as an infinite and a finite square well problem may provide insight into the confinement effect [16, 17, 18]. Band structure engineering (the method of band structure optimization to a particular application) of QW is related to the ability to tune the composition of the heterostructures, hence the electronic and optoelectronic properties. For instance, a parabolic quantum well, $V(z) \sim z^2$, will give rise the eigenvalues, E_n, which are spaced in equal energy intervals. This is a characteristic of a harmonic oscillator. Another extremely important quantum well structure is the triangular well potential, $V(z) \sim z$, which mimics the effect of an applied electric field [19].

If the quantum wells are periodically coupled, then new phenomena may arise because carriers from one quantum well may tunnel through an intervening layer to the next well. The band structure-wavefunctions in SL or MQW structure could be exploited for device construction. If the periodicity and the strength of the interaction between wells are controlled, then there will be possibility of making semiconducting devices with a ‘tailor-made’ band gap, instead of having to rely on the gaps that nature has provided. This will enable devices sensitive to, or operating at, energies in which we are especially interested [15]. In addition, the modulation doping will increase the mobility of the electrons in the QW structures. This feature is exploited in HEMT [1].

Note:

Some more frequently used symbols in the calculations are defined as:

VBO = valance band offset

γ = nearest neighbor interaction

(γ and VBO are in the units of energy that we do not explicitly write in the text)

n = barrier width of QW or sometimes used as quantum number

m = well width of QW

u = central barrier width in double QW

CHAPTER II

THEORETICAL FRAMEWORK

2.1 An overview of the electronic structure problem

The quantum mechanical treatment of materials requires the solution of a complex many-body problem. The Hamiltonian describing the system can be expressed as [20]:

$$H_{total} = \sum_i \frac{\mathrm{p}_i^2}{2m} + \sum_I \frac{\mathrm{P}_I^2}{2M_I} - \sum_I \frac{z_I e^2}{|\mathbf{r} - \mathbf{R}_I|} + \frac{1}{2}\sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{2}\sum_{I \neq J} \frac{z_I z_J e^2}{|\mathbf{R}_I - \mathbf{R}_J|} \qquad (2.1)$$

where indices i, j are for electrons and I, J are for nuclei. The coordinates, momenta and charge of electrons and nuclei are denoted by ($\mathrm{r}_i, \mathrm{p}_i$ and $-\mathrm{e}$) and ($\mathrm{R}_I, \mathrm{P}_I$ and $+z_I\mathrm{e}$) respectively. The mutual interaction of all the constituent electrons and nuclei of the solid determine the actual electronic structure of the system and the properties based on it. In order to solve the Schroedinger equation associated with the Hamiltonian (2.1), two main approximations are usually made: the adiabatic approximation and the one-electron approximation. The adiabatic approximation allows decoupling the dynamics of the fast variables (the electrons) from the dynamics of the slow variables (the nuclei); whereas the one-electron approximation simplifies the complicated many-body problem neglecting the electron-electron interactions [20]. The simplified form of the Hamiltonian of the system can then be written as:

$$H = \sum_i \frac{\mathrm{p}_i^2}{2m} + V(r_i) = \sum_i H_i \qquad (2.2)$$

which denotes for each electron the kinetic energy operator plus the potential energy due to the frozen distribution of nuclei. The resulting Schroedinger equation can hence be solved computing the eigenvalues and the eigenvectors that represent the possible energies and wavefunctions of the electrons in the system.

Many methods have been designed to solve the eigenvalue problem of equation (2.1). Some accurate approaches include the orthogonalized plane wave method, the pseudopotential method, the cellular method, the augmented plane wave method and the Green function method each depending on the choice of a specific formalism to describe the crystal states. Tight binding (TB) approaches are usually less accurate but widely used because of their capability for giving a vivid picture of the electronic structure of several solids, with a modest amount of computational labor [20]. These characteristics are linked to the choice of a simple localized basis to expand the eigenvalue problem.

The TB method of solids as suggested by Bloch in 1928 describes the trends in electronic properties of a variety of systems (periodic or non-periodic) on a rather intuitive basis [21]. The modeling of materials with TB uses a minimum set of parameters and also it is physically transparent. The TB framework is also useful for developing the Anderson and the Hubbard models and it conveniently describes also the electron transport properties [22].

The general TB Hamiltonian (2.2) is often established considering a linear combination of atomic orbitals (LCAO) centered on the nuclei and expanding the Hamiltonian in the subspace $\psi(r) = \sum_j c_j \phi_j (r - R_j)$ and $H_{ij} = < \phi_i \mid H \phi_j >$. The solution of eigenvalue problem $\sum_j H_{ij} c_j = E_i c_i$ provides the energies and the coefficients of the

LCAO for the electronic states. In computing the matrix elements H_{ij} several different schemes may be implemented. Since we are interested in qualitative results we used one orbital per site and neglect orbital overlaps if $i \neq j$ in $< \phi_i | \phi_j > \; = \delta_{ij}$ (orthogonal TB).

2.2 Symmetry property

A transformation that leaves a system unchanged is called a symmetry [23, 24]. The knowledge of the symmetries provides an understanding of several features of the electronic structure of the materials and it is also very useful to simplify the eigenvalue problem used to compute energy band and wavefunctions. In general, the existence of a symmetry property of a quantum system implies the existence of an associated conserved quantity, i.e. conservation of a certain observable associated with the symmetry [25].

Suppose a Hamiltonian *H* commutes with an operator π as $[H, \pi] = 0$ and $| \psi_n >$ is a nondegenerate eigenvector of *H* with eigenvalue E_n; $H | n >= E_n | n >$. The Hamiltonian *H* of a system is said to be invariant under the transformation concerned, and the expectation value of π is a constant of the motion [24]. The invariance of *H* reflects the symmetry of the system with respect to the transformation operated by π. The operator π may represent a continuous (e.g. rotation in atoms) or discrete transformation. For the purpose of this work I will discuss the parity and the lattice translations.

2.2.1 Parity

The transformation on the wave functions under parity (whose eigenvalue must be ± 1) divides wavefunctions into even parity and odd parity as follows: $\pi\psi(-\mathrm{z}) = \pm\psi(\mathrm{z})\begin{cases}\text{even parity}\\ \text{odd parity}\end{cases}$ [24]. Since z and –z are inverse points, equidistant from the origin but in opposite directions, this operation is also called space inversion. We expect the eigenfunctions describing the states of our system to be alternately of even and odd parity provided the potential V(z) is invariant under inversion (V(z) = V(-z)).

2.2.2 Translation

The lattice translation is another kind of discrete symmetry operation, which is applied in a system of regularly spaced positive ions (i.e. periodic potential) [23]. If we limit our discussion to the one-dimensional case, then the potential is such that $V(z) = V(z + \tau)$; where $\tau(= \mathrm{na})$ is the lattice translation, and '*a*' the interatomic distance with n an arbitrary integer. For crystals, the Hamiltonian has translation symmetry. For each τ we can define a translation operator such that $[T_\tau, H] = 0$. This implies the energy eigenstates of *H* are the simultaneous eigenstates of all the T_τ [21]. We can choose the eigenstates $\psi(z)$ of *H* so that for every τ ,

$$T_\tau\psi(z) = \psi(z+\tau) = c(\tau)\psi(z) = e^{ik\tau}\psi(z)\,, \tag{2.3}$$

which can only hold if

$$\psi(z) = e^{ikz}u_K(z) \tag{2.4}$$

with u_K a periodic function (or modulating function which have the lattice periodicity)

$$u_K(z) = u_K(z + \tau) \tag{2.5}$$

This is precisely the Bloch's theorem (or Floquet's theorem in 1-D) as stated: the wavefunctions of the crystal Hamiltonian can be written as the product of a plane wave of wavevector k, times an appropriate periodic function. The Bloch theorem justifies the use of k as a quantum number and allows to write $E_{n,k}$.

Moreover, the symmetry properties are useful to reduce the computational cost in determining energies. For each energy level $E_{n,k}$ the translational symmetry implies $E_{n,k} = E_{n,k+G}$ where G is a vectors that defines the reciprocal lattice [26]. Only those k whose values are not related by a G have to be considered; and their set may be used to define a set of independent k called the first Brillouin zone (BZ) [27].

2.3 Electronic band structure in a periodic potential and effective mass

The concept of the Bloch function is at the core of the electronic band structure for a periodic system. When the Bloch function of the form $\psi_{nk}(z) = \frac{1}{\sqrt{N}} e^{ikz} u_{nk}(z)$, where N is the number of unit cells in the crystal, is substituted in Schroedinger equation, it yields an eigenvalue equation in which both eigenvalues and eigenfunctions depend on k. The new eigenvalue equation leads us to a large number of solutions, giving a set of energies $E_{1,\mathrm{k}}$, $E_{2,\mathrm{k}}$, … for each value of k. From the periodic boundary conditions applied to a macroscopic crystal it results that k is dense in the limit of $N \to \infty$. We can then write the energy eigenvalue as $E_{\mathrm{n}}(\mathrm{k})$, where the subscript *n* as the band index indicates the band to which the particular state belongs. The number of possible wavefunctions in a band is given by the number of distinct wave vector, k, in the first BZ. Each band has *N* states, and can be occupied by 2*N* electrons [10, 26].

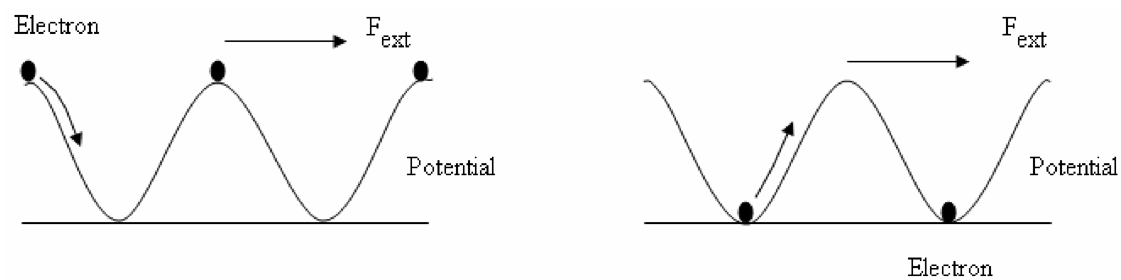
Ideally the number of bands is large but only the ones below the Fermi level are occupied by electrons. The energy intervals between any two successive bands constitute the energy gaps, which are forbidden energies where there are no states for electrons to occupy. Properties such as optical absorption and conductivity depend upon the magnitude of the gap between occupied and unoccupied states.

The electron in a crystal is sometimes known as Bloch electron to appreciate its unusual properties as compared to a free electron. Besides an external field, the electron experiences the crystal field. This will cause a change in the dynamics of the electron compared to a free electron. The electron can accelerate more rapidly in the solid than in free space due to effective mass effects in an energy band. The effective mass [Appendix G] is given by the relation: $m^* = \frac{\hbar^2}{\partial^2 E / \partial k^2}$. The mass m^* is inversely proportional to the curvature ($\partial^2 E / \partial k^2$) of the band. Near the bottom of the band m^* has constant value and usually $m^* > 0$. As k increases m^* is no longer a strict constant, and beyond the inflection point of the band $m^* < 0$, i.e. the electron has negative effective mass of electron closer to the top of the band [26].

The physical meaning of the effective mass can be interpreted conveniently by considering the crystal momentum, $\hbar\mathrm{k}$, in the semiclassical model. A Bloch electron in the state ψ_k behaves as if it had a momentum $\hbar\mathrm{k}$, because external force: $\frac{d(\hbar\mathrm{k})}{\mathrm{dt}} = \mathrm{F}_{\mathrm{ext}}$. However, $\hbar\mathrm{k}$ is not equal to the actual momentum of the Bloch electron. This indicates that there exists a force (F_L) exerted by the crystal lattice on the electron in addition to external force. The effective mass in such situation can be deduced as:

$$m^* = m_0 \frac{F_{ext}}{F_{ext} + F_L} \tag{2.6}$$

where m_0 is the free electron mass. m^* may be smaller or larger than m_0 or even negative, depending on the lattice force, F_L. In order to be more explicit about m^* in the crystal let us suppose the Figures 2.1 (a) and (b) the electrons piling up initially near the top and the bottom of the crystal potential respectively.



(a) (b)

Figure 2.1 Energy bands leading to (a) $m^* < m_0$ (b) $m^* > m_0$

When an external force is applied in Figure 2.1 (a), the electron rolls downhill along the potential curve resulting in a positive F_L getting operative and hence $m^* < m_0$. On the other hand in Figure 2.1 (b), the F_L tends to oppose the F_{ext} which results in $m^* > m_0$. If the crystal potential is sufficiently steep, then F_L becomes larger than F_{ext} and m^* becomes negative [26].

2.4 Basic approaches for heterostructures

2.4.1 Confinement potentials

In a QW, where different materials are combined to form a heterostructure, the shift in the band structure induces a step like potential profile in the growth direction as shown in Figure 2.2. The height of the discontinuity (in the so called abrupt junction approximation) is called band offsets. The conduction band offset confines the electrons whereas the valence band offset confines the holes.

At present it is possible to realize different shapes of confining potentials due to advancements in growth techniques. The simplest and most commonly employed configurations are the single or multiple square quantum wells. The degree of confinement as well as the degree of overlap between confined electronic states can be controlled by the width and depth of the confining potential well, and the width of the barrier layer. Although the square well configuration in the form of single, multiple or SL structure remains the most commonly studied, techniques such as MBE permit realization of other functional forms of the potential through compositional grading of the barrier and well layers. Some of those potential forms of quantum well are parabolic, triangular, trapezoidal, V-shaped etc. Any such potential shape may create the new electronic quantum systems which may be exploited for novel devices [19].

2.4.2 Quantum well and confined states

When the Fermi energies through three layers of material 1/material 2 multilayer structure equalize, then the schematic of quantum well will be as shown in the Figure 2.2. Material 1 and material 2 represent ‘well’ and ‘barrier’ respectively [28]. Experimentally, QW are formed by sandwiching a narrower band gap crystal or semiconductor (the well) with the wider band gap crystals as adjacent layers (the barriers). The relative heights of the discontinuities in the conduction band (CB) and valence band (VB) edges are intrinsic properties of the two materials involved. Since the narrower band gap material forms a potential well in the CB and VB, the electrons and holes are bound states in the growth direction (z-direction), while their motion in x-y plane is unrestricted. This implies the carriers are quantized in one-dimension, i.e. in the direction perpendicular to the interfaces. This is known as *quantum confinement*. Thus, a one-dimensional confinement effect occurs in QW which is used to engineer the electronic properties of the materials [9]. The chemical composition, dimension of the layers, and the band gap energies vary the properties of QW. The confinement effect may be increased by nanostructuring as quantum wire and quantum dot [Appendix F].

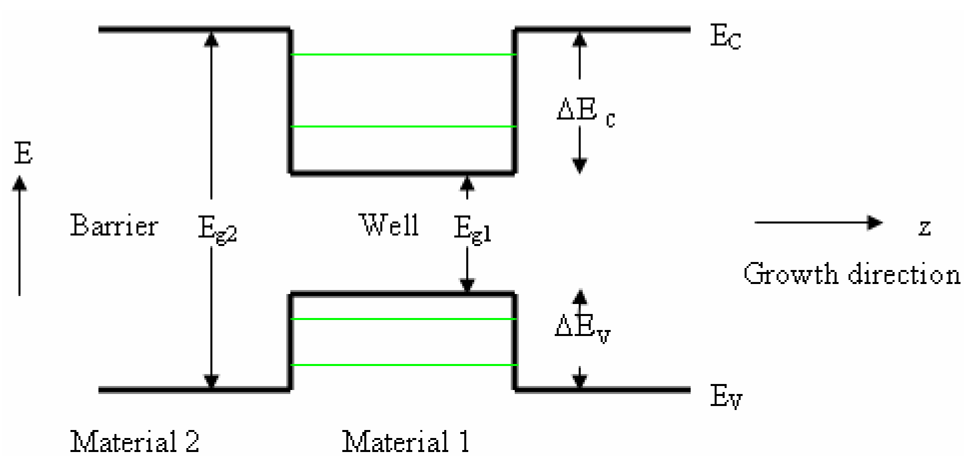


Figure 2.2 Schematic of quantum well as a function of distance in the growth direction z and energies of the sub-bands within the well

The width of the quantum well determines whether the confinement effect will be important. Usually, the carriers experience significant quantum confinement effects for well widths of order 100 Å or less. It is reasonable to consider the 'well' width be thick enough (L > 20 Å) so that the well layer material has the properties characteristic of macroscopic crystal and thin enough (L < 400 Å) so that the width 'L' is shorter than the mean free path ($\lambda = v\tau\,;\tau = m^*/\rho ne^2, \rho$ is resistivity), i.e. an electron is likely to traverse the well without undergoing a collision of any kind [9, 29].

2.4.3 Band structure and density of states of quantum well

The confinement of the electron's wavefunctions in reduced dimensions dramatically modifies the electron energy spectrum [30]. The density of state (DOS) diagram as given in the Figure 2.3 shows the patterns of energy spectrum in QW and compares it with the bulk material [31].

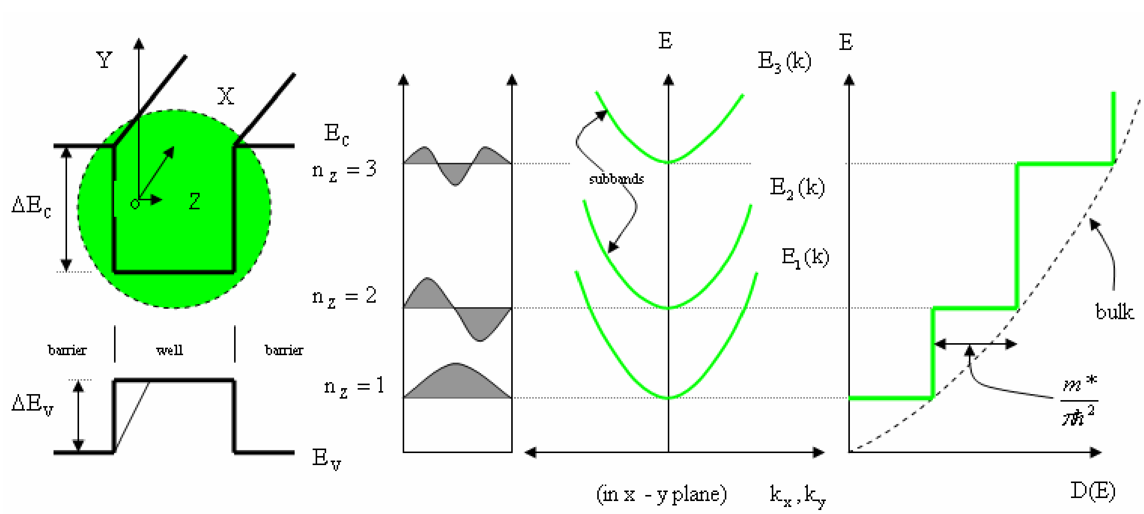


Figure 2.3 Energy levels, dispersion curve and DOS of quantum well. The dashed curve is DOS of bulk semiconductor material

If we assume an infinite confining potential, i.e. $V(z) = \begin{cases} 0 & |z| \le \dfrac{L}{2} \\ \infty & \text{elsewhere} \end{cases}$, then the dispersion relation in each isotropic subband n and the corresponding wavefunctions can be written as [31]

$$E_n = E_n(z) + E(k_x, k_y) \equiv \frac{\pi^2 \hbar^2}{2m^* L_z^2} n^2 + \frac{\hbar^2}{2m^*}(k_x^2 + k_y^2) \tag{2.7}$$

and

$$\psi = \phi_n(z) \exp(ik_x x + ik_y y) \tag{2.8}$$

respectively, where n = 1, 2, … the quantum confinement numbers, L_z is the confining dimension, k_x and k_y are the components of wavevector along the transverse directions, m^* is the carrier effective mass, and $\phi_n(z) = \sqrt{\dfrac{2}{L_z}} \sin(\dfrac{\pi n}{L_z} z)$. The term $\exp(ik_x x + ik_y y)$ describes the electronic motion in directions x and y, similar to free electrons wavefunctions. Also, note that: $L_z = n\dfrac{\lambda}{2}$ leads to $k_z = \dfrac{2\pi}{\lambda} = \dfrac{\pi}{L_z} n$. For n confined states within the quantum well system, the DOS at any particular energy is the sum over all sub-bands below that point. Thus, the DOS per unit area can be expressed as [14]:

$$D(E) = \frac{m^*}{\pi\hbar^2} \sum_n \theta(E - E_n) \tag{2.9}$$

where $\theta(E - E_n)$ is the unit step function (staircase DOS). The total energy of an electron in the well is given by each value of E_z plus the permitted $(E_x + E_y)$ contributions, so the DOS associated with $(E_x + E_y)$ is constant, i.e., the DOS has the same value at the band edge as for higher energies due to the 2D character of carrier motion [15]. Therefore, the DOS of quantum well has a step-like form, each step originating from the onset of a sub-band. This implies that the DOS at the bottom of the quantum well system is finite, whereas that for bulk crystal is zero. The DOS of a bulk parabolic band is $\sim\sqrt{E}$, as indicated by the broken line in Figure 2.3. As the well width becomes thicker the steps become smaller and more closely spaced and the DOS starts to approximate to the parabolic shape of the bulk crystal DOS [9].

The electrons' motion in the z-direction is quantized into subbands; however, the x-y motion is unconstrained, so that a degenerate 2-D electron system results [10]. The charge carriers concentrate near the band edges due to the staircase DOS of QW. This improves the characteristics of a QW, particularly opto-electronic interaction enhancement. This has led to a lower threshold current and higher characteristic temperature in quantum lasers and higher nonlinear coefficients for the realization of optical bistable devices (OBD) [28]. DOS function is important, in the electronic processes, particularly in transport phenomena.

2.4.4 Envelope function approach to heterostructures' electronic problem

The formal justification of the QW treatment in 2.4.3 is based on the envelope function approximation (EFA). Since the crystallographic (e.g. lattice constants) and chemical properties of the layered materials of heterostructures are very similar or almost matching, the rapidly-oscillating part of the Bloch function (the part which has the periodicity of the lattice) is the same [32]. Because of this reason, instead of considering the full wavefunction only the modulation (envelope) of the constituents Bloch wave is retained [33]. This shows the effects induced by the heterostructure potential profile. The physical properties of the heterostructures can be derived from the slowly varying envelope function, which is identified as $\psi(z)$, rather than the total wave function $\psi(z)u(z)$ where $u(z)$ is rapidly varying on the scale of the crystal lattice. An important point is that the boundary conditions should include the effective mass since $m^* = m^*(z)$.

Let the envelope-functions in 'well' and 'barrier' be $\phi_{\mathrm{W}}(z)$ and $\phi_B(z)$ respectively, of a quantum well whose well width is 'L', then the boundary conditions $\phi_{\mathrm{W}} = \phi_B$ and $\frac{1}{m_W^*}\frac{\mathrm{d}\phi_W}{\mathrm{dz}} = \frac{1}{m_B^*}\frac{\mathrm{d}\phi_B}{\mathrm{dz}}$ imposed at interfaces (i.e. $\mathrm{z} = \pm\frac{L}{2}$) will produce two transcendental equations [10], $\tan(\frac{ka}{2}) = +\frac{m_W^*\kappa}{m_B^*k}$ and $\cot(\frac{ka}{2}) = -\frac{m_W^*\kappa}{m_B^*k}$, whose solutions give bound states in the well; where, $\kappa = \sqrt{\frac{2m_W^*\mathrm{E}}{\hbar^2}}$ and $\mathrm{k} = \sqrt{\frac{2m_B^*(\mathrm{V_o} - \mathrm{E})}{\hbar^2}}$, provided $\mathrm{E} < \mathrm{V}_o$, $\mathrm{V(z)} = 0$ at bottom of the well and V_o at the top. This problem can only be solved numerically or graphically, where $\mathrm{V}_o \equiv \Delta\mathrm{E_C}$ *or* $\Delta\mathrm{E_V}$ [Appendix C]. We tackle this problem by constructing parameterized tight-binding Hamiltonian of the system.

2.5 Tight binding model

The basis of the tight binding band approximation is that the outer electrons are assumed initially to be closely associated with their atoms; that is, we consider a large atomic potential so that the electrons remain mostly bound to the ionic cores [41].

In order to explore the binding of the atoms in a solid or the electronic properties of a 1-D heterostructure we assume first a one-dimensional crystal, formed by bringing individual atoms close together and centered in the lattice positions $\tau_n = na$. We also assume one atomic orbital per site, ϕ_n, and a one electron Hamiltonian. An additional assumption of our TB model is that the function ϕ_n decays rapidly and is almost negligible by the time it reaches to neighboring site at ϕ_{n-1} or ϕ_{n+1}. The TB Hamiltonian is built starting from a Bloch function $\psi_n(z) = \frac{1}{\sqrt{N}} \sum_n \phi_n e^{ik\tau_n}$ such that

$$H_{nm} = <\psi_n | H | \psi_m> = \sum <\phi_n | H | \phi_m> e^{ik(\tau_m - \tau_n)}. \tag{2.10}$$

Imposing the nearest neighbors (NN) approximation, we get, $<\phi_n | H | \phi_m> = 0 \quad \forall \quad |n-m| \geq 2$; so that

$$H_{nm} = <\phi_n | H | \phi_m> (\delta_{nm} + \delta_{n+1,m} e^{ika} + \delta_{n-1,m} e^{-ika})$$

Assuming $<\phi_n | H | \phi_m> \ = \varepsilon$ and $<\phi_n | H | \phi_{n\pm1}> \ = \gamma$ the energy becomes

$$E(k) = E_0 + 2\gamma \cos(ka) \tag{2.11}$$

where E corresponds to the on-site (OS) diagonal term and γ is called hopping parameter. The tri-diagonal Hamiltonian (TDH) as shown in Figure 2.4 provides a pictorial representation of a one-dimensional crystal described with tight-binding model [20].

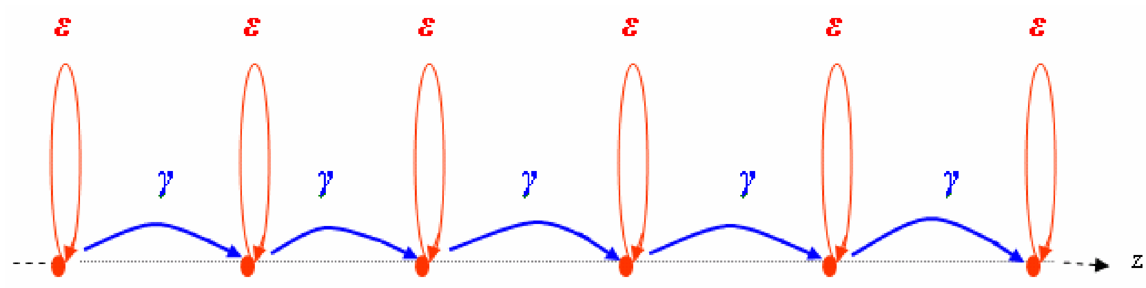


Figure 2.4 Schematic representation of TDH. Diagonal matrix elements and off-diagonal hopping integrals are denoted by ε and γ respectively

When the next nearest neighbor (NNN) interaction is considered in the crystal, the band structure relation becomes $E(k) = E_0 + 2\gamma_1 \cos(ka) + 2\gamma_2 \cos(2ka)$, where γ_1 and γ_2 are NN and next NN (NNN) interactions respectively. The dispersion curve E(k) is,

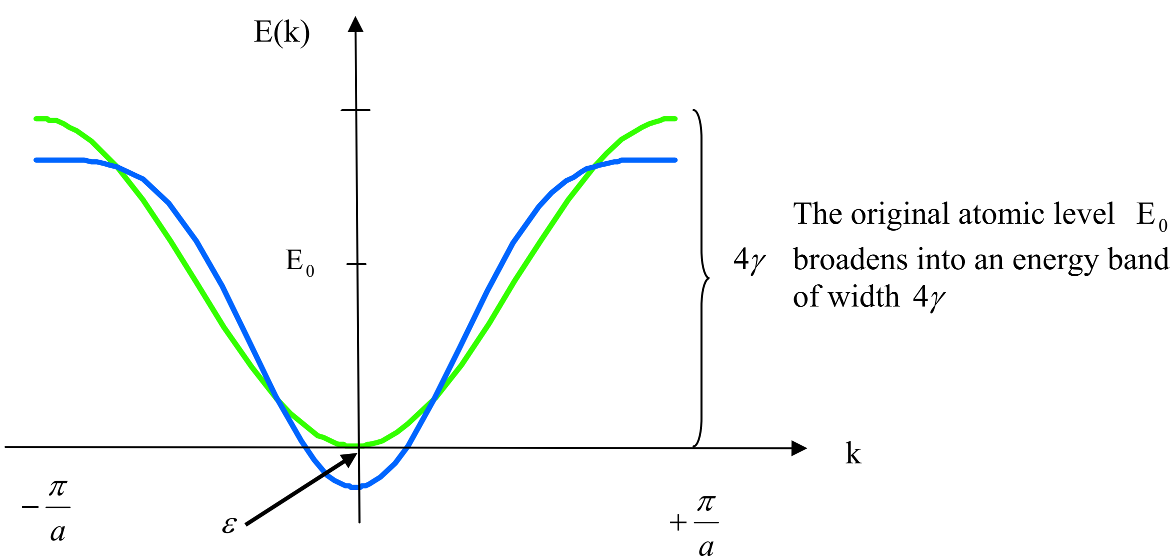


Figure 2.5 Dispersion E(k) as a function of wave vector k: NN band (Green) and NNN band (Blue)

2.5.1 Notes about the tight-binding bands

1. We see that even tightly bound atomic states form bands in crystals. The γ describes the hopping integral or transfer integral: it is the measure of strength of an electron to transfer from one atom to the next. This also gives a direct measure of the bandwidth (energy width). Since the atomic functions are localized, the overlap between them decreases sharply with increasing distance. The greater the overlap, the stronger the interaction and consequently the wider the band is.

2. The Figure 2.6 shows the origin of the bands (2.11). This also gives a new interesting feature: as the nearest neighbor separation a decreases the NN wavefunctions' overlap increases and the band widths 4γ, $4\gamma^{'}$ increase. At a certain critical separation, adjacent bands overlap. This destroys the forbidden energy gap between the two bands; this describes the insulator to metal transition at high pressure.

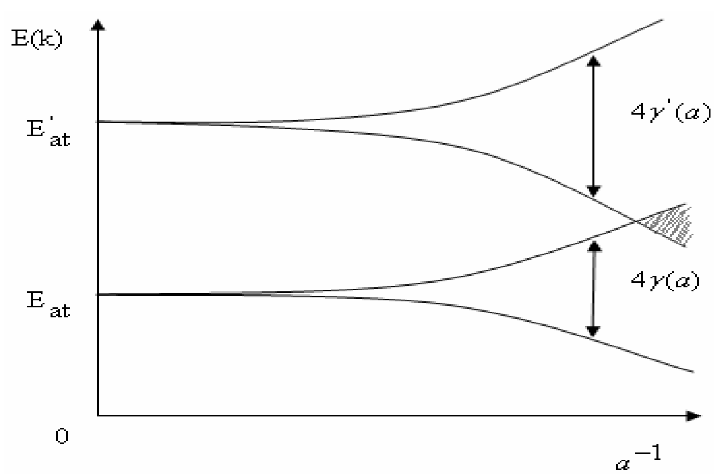


Figure 2.6 A general feature of energy of an electron in TB model [9]

3. Near the bottom of the band where k is very small so the approximation can be made as:

$$E(k) = E_0 + 2\gamma\cos(ka) = E_0 + 2\gamma - 2\gamma\frac{k^2a^2}{2} = E_0^{'} - \gamma k^2a^2$$
$$E(k) - E_0^{'} \approx -\gamma k^2a^2 \equiv \frac{\hbar^2k^2}{2m^*} \Rightarrow \text{FE energy.}$$

This implies $m^* = -\frac{\hbar^2}{2\gamma a^2}$ for the TB band at the free band zone (FBZ) center where γ is assumed to be negative. For $\gamma > 0$, the dispersion curve just flips. The small γ leads to large effective mass and narrow bandwidth, indicating that it is hard to move the electrons around; large γ leads to light effective mass and large bandwidth, indicating that it is easy to move the electrons around. This connects with the real-space picture of the formation of TB bands, in which the transfer integrals reflect the ease with which an electron can transfer from atom to atom. In the case of NNN TB the effective mass at FBZ center is $m^* = -\frac{\hbar^2}{2(\gamma_1 - 4\gamma_2)a^2}$.

2.5.2 Modeling of tight-binding problems

The mathematical representation of the eigenvale problem is set up by constructing a tri-diagonal Hamiltonian matrix. The elements are different from zero only on the principal diagonal representing on-sites (OS) and the next upper and the next lower ones representing the nearest neighbor (NN).

The stationary Schroedinger's equation for one-dimensional system is

$$\mathrm{H}\Psi(z) = \mathrm{E}\Psi(z). \tag{2.12}$$

Here,
$$H = \begin{pmatrix} \varepsilon_0 & \gamma & . & . & \gamma e^{ikL} \\ \gamma & \varepsilon_0 & \gamma & . & . \\ . & \gamma & \varepsilon_0 & \gamma & . \\ . & . & . & . & . \\ \gamma e^{-ikL} & . & . & . & . \end{pmatrix} \qquad (2.13)$$

for the case of periodic boundary condition (PBC). The PBC imposed on the problem introduces the k dependence in the Hamiltonian. The numerical energy solution (i.e. TB band) for a finite linear chain (without PBC) as shown in Figure 2.7 (Blue curve) perfectly reproduces the analytical TB band structure of equation (2.11). Since the chain is very long we are able to reproduce the expected TB band. The band fits the free electron dispersion (Red curve) when TB effective mass ($m^* = -\frac{\hbar^2}{2\gamma a^2}$) is considered in the problem. In this case the energies are given by

$$E(n) = \gamma \frac{\pi^2}{N^2} n^2 \qquad (2.14)$$

and this represents the expected quadratic dispersion with $k_Z = \frac{\pi}{\mathrm{N}} n$ (N is the number of sites present on the linear chain).

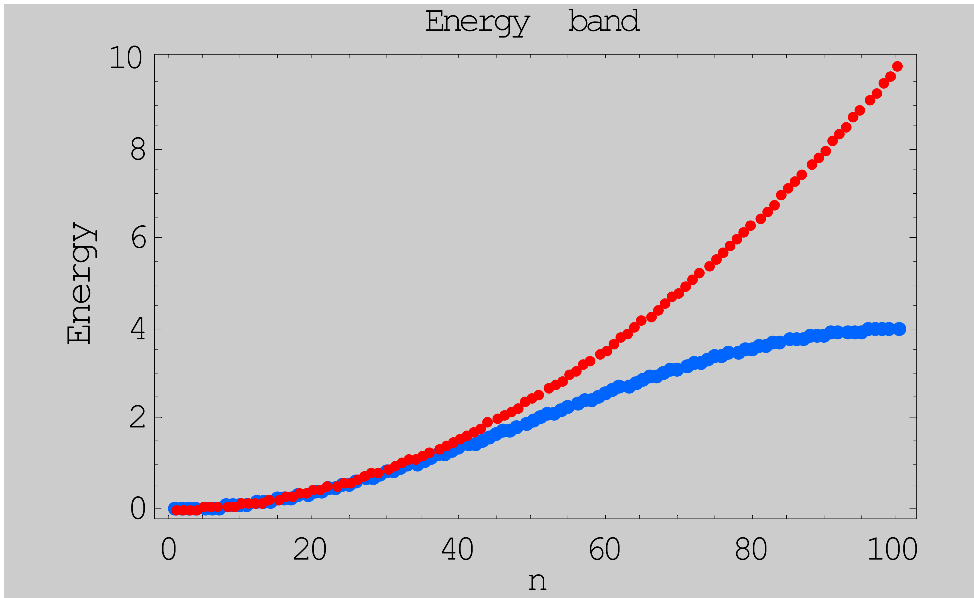


Figure 2.7 Numerical solution of particle in a box (PIB) problem: Blue (direct calculation), Red (TB effective mass correction). Parameters: n = 100, OS energy = 2.0, γ = 1.0

The deviation from the quadratic dependence at higher n is intrinsic to the TB model that uses a discrete basis set.

If we do not impose the PBC but try to approach the extended system using larger and larger H (a large number of sites) we fail to reproduce the correct wavefunctions. Indeed our problem is a representation of a particle in a box (PIB) i.e. infinite barrier QW [34]. The numerical wavefunctions of the problem are given in Figure 2.8 below.

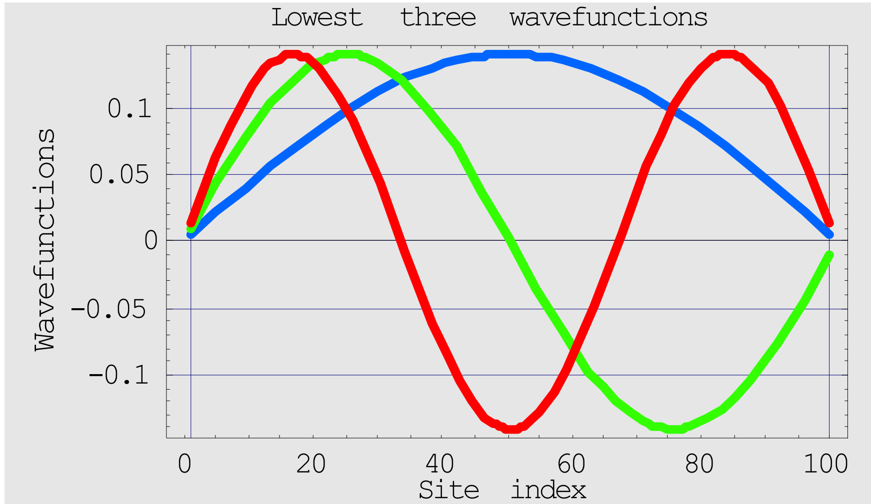


Figure 2.8 Numerical solutions of PIB problem: wavefunctions for the first three stationary states: Blue (n = 1), Green (n = 2) and Red (n = 3). Problem construction parameters: n = 100, OS energy = 2.0, γ = 1.0

As can be seen in the Figures 2.7 and 2.8, numerical solutions match the analytical solutions, $E_n = \frac{\pi^2 \hbar^2}{2mL^2} n^2$ and $\psi_n(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi z}{L}\right)$ obtained by plugging in boundary conditions in the solutions of Schroedinger equation, of the *infinite flat potential well* problem [35]. We can remark in the result that the n^{th} eigenstate possesses (n-1) zeroes in addition to the zeroes at the extremities as in the infinite flat potential well; this is expected from orthogonality considerations. The wavefunctions are eigenstates of the parity.

We know, $E_n \propto \frac{n^2}{L^2}$ analytically, and this is calculated in Figure 2.9 (continuous curves). However, the Figure 2.9 (dots) displays the numerical results of calculations of the three lowest energy states of an electron in the hypothetical well (HW) with infinite barriers. All three states show the same monotonic behavior, with the energy decreasing as the well width increases.

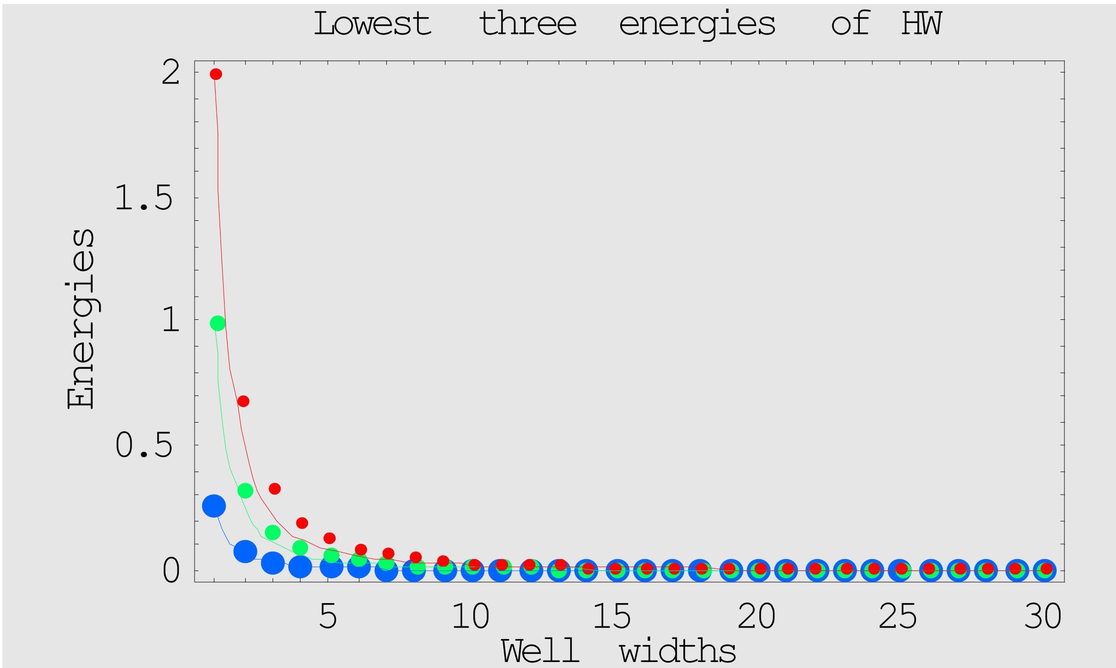
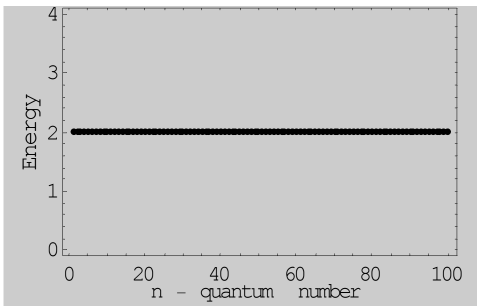
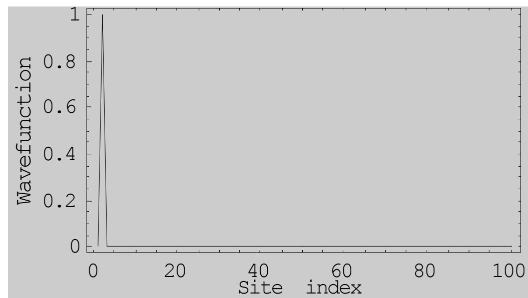


Figure 2.9 Continuous curve: analytical results, and dots: numerical results. Blue (1st confined level), Green (2nd confined level) and Red (3rd confined level)

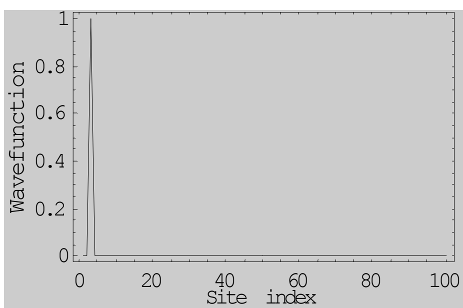
The problem can be treated with the idea of *band spectrum and delocalization*. When the atoms are infinitely far apart ($a = \infty$), the individual bound states of an electron within each atom are all identical [35]. These atom states are N in number. They are both *stationary* and *localized*. The corresponding energy levels form a discrete spectrum; each level is N-*fold degenerate* since it corresponds to N states localized at different sites. This is investigated in the program with $\gamma = 0$ (NN interaction cancelled) which is now a case of isolated atoms. The isolated atoms of the same kind have the same eigenvalues throughout the chain, and the wavefunctions at the corresponding sites are delta-like functions. This behavior is investigated in computation as shown in the Figures 2.10 (a), the constant eigenvalues in the energy band, and ((b), (c) & (c)) the peaks of the delta-like wavefunctions for the first two states.



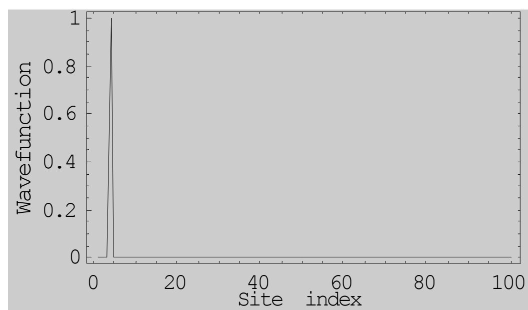
(a)



(b)



(c)



(d)

Figure 2.10 (a) Constant eigenvalues; (b), (c) & (d) Delta-like eigenfunctions

The band structure obtained by solving numerically the eigenvalue problem with the periodic boundary condition in one-dimensional chain of crystal is given in the Figure 2.11 below:

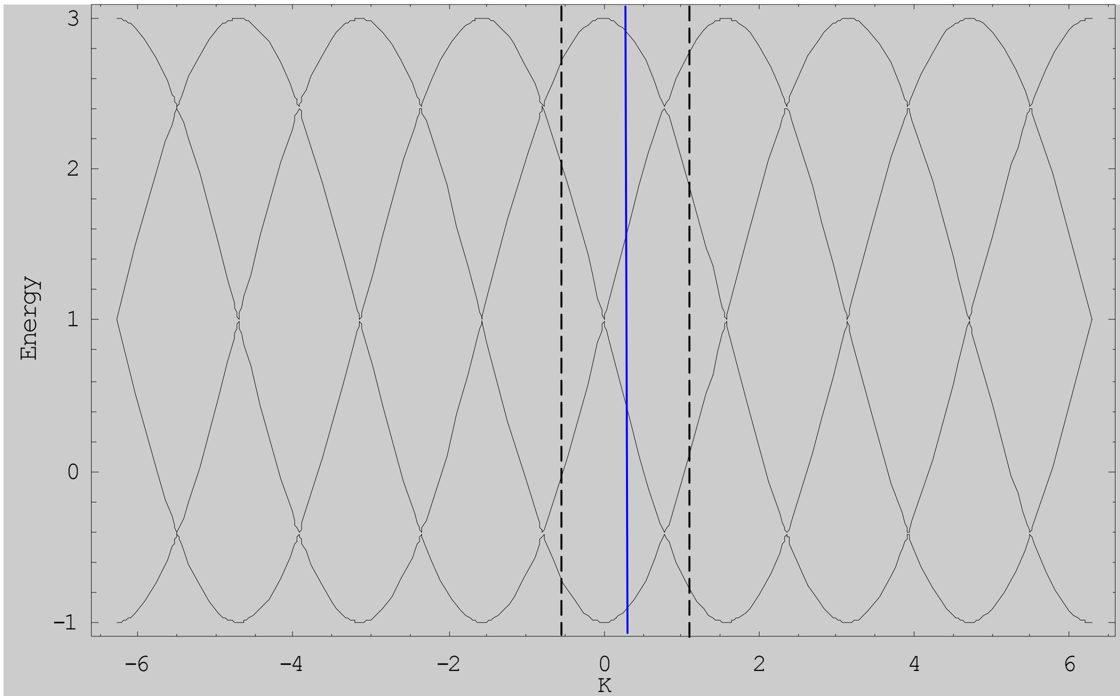


Figure 2.11 The band structure of one-dimensional chain of crystal with periodic boundary condition

The band structure around $k \approx 0$ indicates that the periodic boundary condition in the one-dimensional chain of crystal makes the system similar to the free electron model, for which the eigenvalues come in $E(k) \cong \dfrac{\hbar^2 k^2}{2m}$.

CAPTER III

RESULTS AND DISCUSSIONS

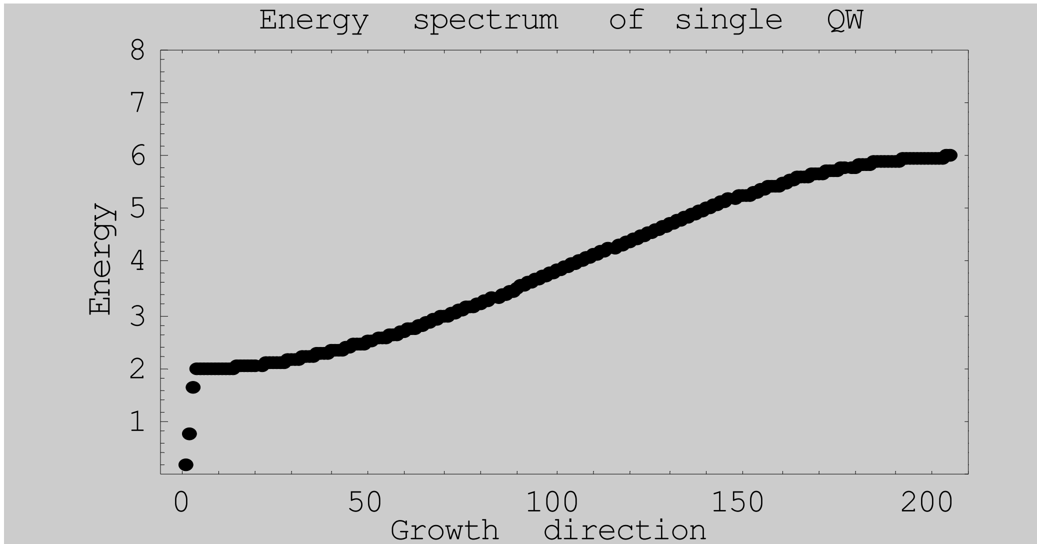
Tight binding calculations provide an effective way to explore quantum effects in heterostructure physics [36]. I present my results obtained with Mathematica [Appendices A and B] in the following paragraphs.

3.1 Isolated single quantum well

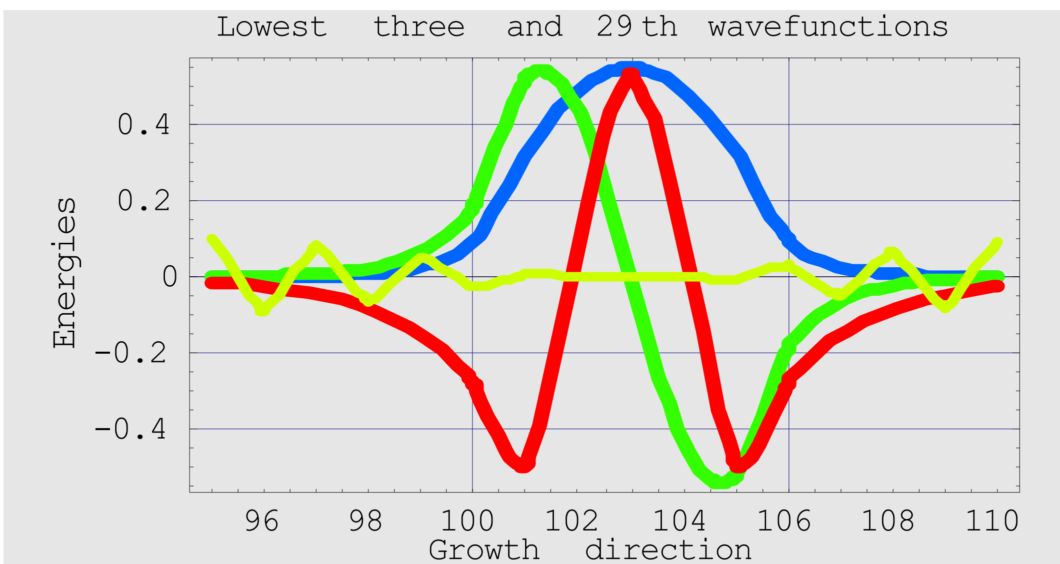
Although the infinitely deep confining potential provides interesting insight for understanding the physics of a QW, it can not account for features like barrier height and width or for unusual boundary conditions. The finite QW model is more relevant from a fabrication point of view of semiconductor heterostructures [37]. For example, a layer of GaAs sandwiched between two thick layers of $Ga_{1-x}Al_xAs$ would form a Type-I quantum well with finite band offsets [38]. If the TB matrix elements are tuned to reproduce the bulk systems, then a good description of the QW may be expected with the addition of only one parameter: the valence band offset (VBO).

In a QW the energy spectrum consists of a finite sequence of discrete values up-to the VBO, and at higher energy it is continuous. We can verify that even for a very shallow QW, i.e. very small VBO, there always exists at least one bound state. This feature is particular to one-dimensional systems. Numerical results in the Figure 3.1(a) clearly show three confined levels (discrete dots) and unbounded states (a continuous curve) in an isolated finite quantum well for a particular set of well depth and well width. For a bond length ‘a’ the well width is ‘ma’ and the boundary conditions are applied at z

= 0 and z = L where L = (2n + m)a for n and are the number of atoms in barrier and well respectively.



(a)

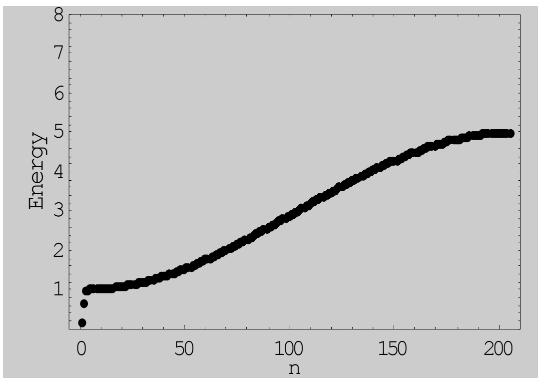


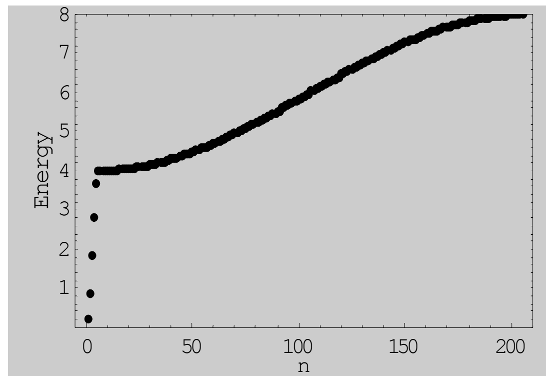
(b)

Figure 3.1 QW (a) eigenspectrum (b) first three eigenstates (confined) and 29th eigenstate (unconfined): Blue (n = 1), Green (n = 2), Red (n = 3), Yellow (n = 29). Problem construction parameters: VBO = -2.0, γ = -1.0, barrier width =100, well width = 5

The Figure 3.1(b) shows that the first confined state wavefunction is, as expected, nodeless and the higher levels have 1, 2, etc. nodes (in general n-1 nodes for nth state). Inside the well, the wavefunctions resemble the solution of the infinite-barrier problem. However, their amplitudes at the well boundaries are *finite*, and decay exponentially in the barrier. Such characteristics of wavefunctions demonstrate why envelope functions such as $\cos(qx)$ and $\sin(qx)$ are appropriated for the well region and $e^{\pm qx}$ for the barrier region [10]. Also, it is evident from the wavefuctions result that the higher the energy the longer the exponential tails, i.e. the greater the leakage of wavefunctions. The leakage of wavefunctions outside the QW wall is a consequence of quantum mechanical tunneling of electrons. In other words, the spatial confinement is the strongest (i.e. the smallest tunneling) for the first confined state and it gradually decreases (tunneling gradually increases) as we go to higher energy levels. The eigenfunctions above the highest confined level (n = 3 is the highest confined level in our example, Figure 3.1(a)) of QW are not strictly localized because they have so high energy that they are less affected by the potential of the well. The eigenfunction for n = 29 in Figure 3.1 (b) shows the limit of our TB scheme in describing states with more nodes than sites in the well.

The number of confined levels in QW depends on depths (i.e. VBO or CBO) of the well as shown in the Figure 3.2. The depths of the well for bands in Figure 3.2(a) and 3.2(b) are 1.0 and 4.0 units, respectively, and in Figure 3.1 (a) this is taken as 2.0 units. Hence the numbers of confined levels are 3, 2 and 5 in Figures 3.1 (a), 3.2 (a) and 3.2 (b) respectively. Provided all other QW parameters remain constant, the number of confined levels increases with the increase in well depth gradually only up-to a certain limiting depth. Beyond this limit the number of confined level remains constant even if the depth is increased. This is a consequence of the TB approach and its discrete basis. A typical numerical result is given in the Figure 3.3 to show this feature. However, each level shifts slightly higher than previous position as the depth of well increases whether the number of confined levels is changed or not, which is evident from the result given in Figure 3.4.





(a) (b)

Figure 3.2 Energy spectrums for (a) VBO = -1.0 and (b) VBO = -4.0; for n = 100, m = 5, γ = -1.0

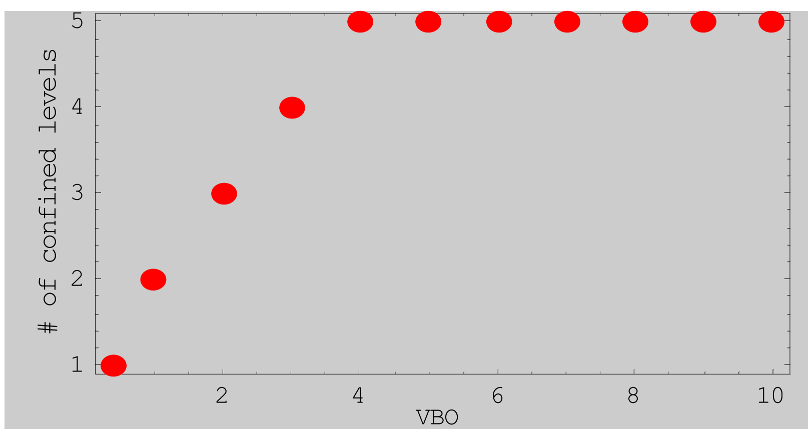


Figure 3.3 Number of confined levels as a function of VBO keeping other QW parameters constant as: n = 100, m = 5, γ = -1.0

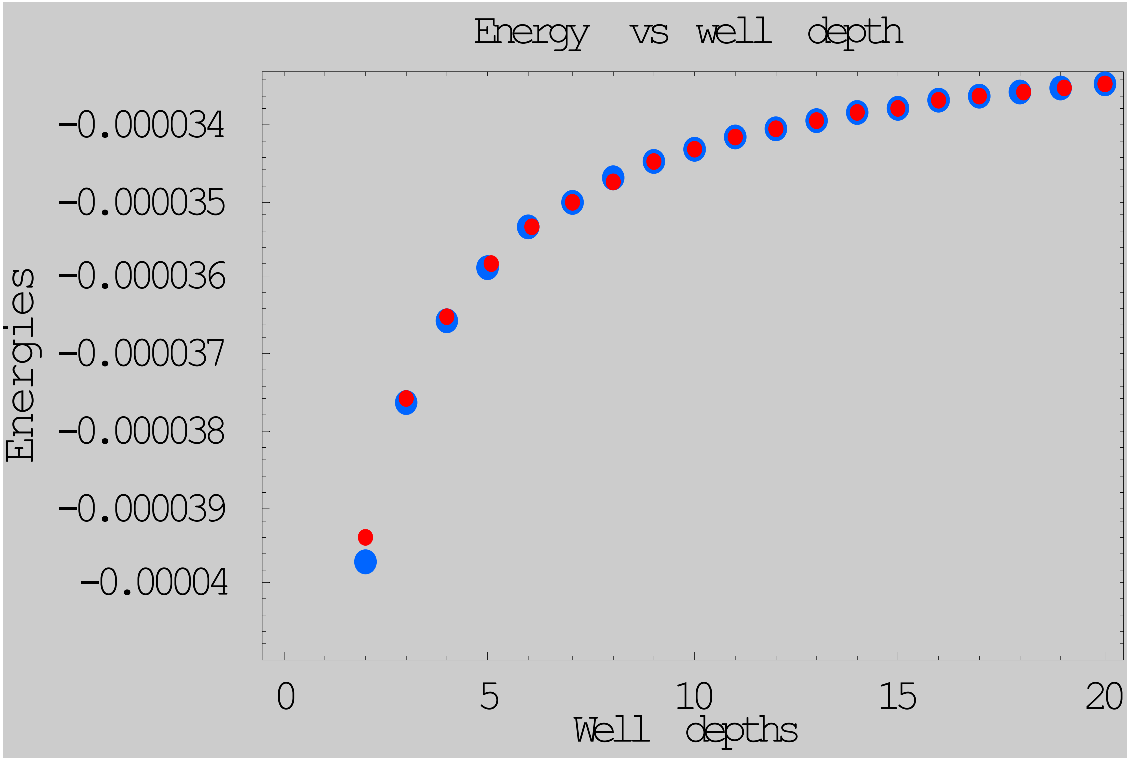


Figure 3.4 Lowest two confined levels as a function of VBO keeping other QW parameters constant as: n = 100, m = 4, γ = -1.0; blue circles (first confined level) and red circles (second confined level)

The number of confined levels in QW also depends on widths of the well directly. The table 3.1 shows the number of confined levels for different widths at constant VBO. The number of confined levels increases as the well widths increase. It is noticed that the energy levels are less confined when the QW width increases as expected by the relation $E_n \approx \frac{n^2}{L^2}$ (∞ barrier QW case).

Table 3.1 Number of confined levels as a function of well widths for QW with

n = 100, m = 5, γ = -1.0 and VBO = -2.0 (see introduction for definition of notations)

Well Width	Number of confined levels	Energies of confined levels				
		E(1)	E(2)	E(3)	E(4)	E(5)
1	1	1.1715700	-	-	-	-
2	1	0.6666670	-	-	-	-
3	2	0.4246150	1.500000	-	-	-
4	2	0.2928930	1.082010	-	-	-
5	3	0.2138350	0.808512	1.639460	-	-
6	3	0.1628220	0.624382	1.304520	-	-
7	4	0.1280470	0.495630	1.053830	1.71686	-
8	4	0.1033030	0.402466	0.865788	1.44057	-
9	5	0.0850801	0.333045	0.722419	1.21804	1.76628
10	5	0.0712776	0.280011	0.611126	1.04005	1.53197

The numerical result shown in Figure 3.5 indicates that as the well width increases the energy decreases monotonically for all three confined levels of the QW. The contribution to the electron energy due to confinement effect decreases and the total energy for the electron tends towards the bulk limit.

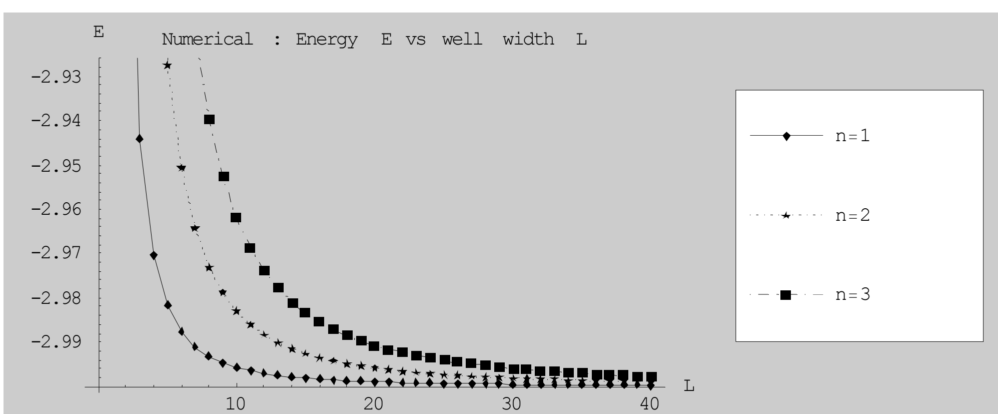


Figure 3.5 Energy as a function of well widths for first, second and third confined levels of a QW

3.2 The double quantum well

The TB method to solve the Schroedinger equation can be applied to systems for which the analytical solutions are difficult or impossible. A double quantum well is solved on the basis of this notion to describe the electronic structure features that apply also to bi-atomic molecule. The Figure 3.6 displays the lowest two confined energy states as a function of the central barrier width for a symmetric double quantum well with a fixed arbitrary well width.

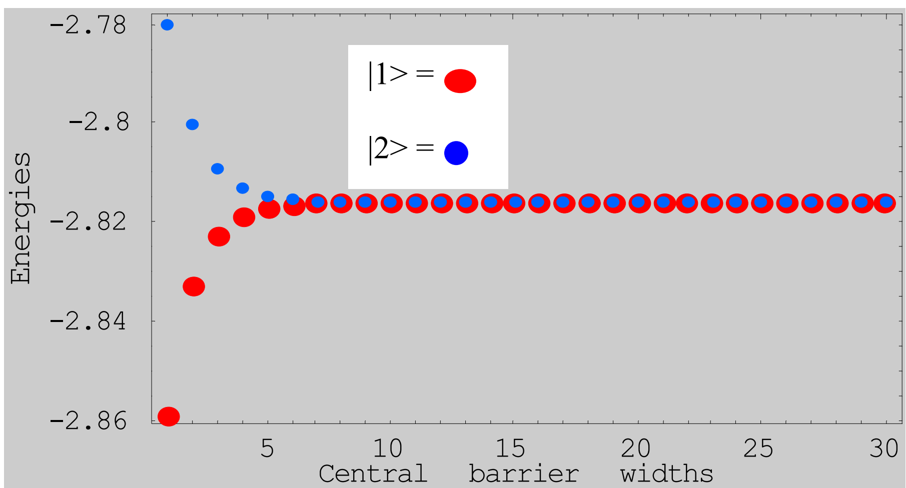


Figure 3.6 The lowest two confined energies of the symmetric QW as a function of the central barrier width. Parameters: n = 10, m = 5, γ = -1.0, VBO = -1.0 (see introduction for definition of notations)

When the wells are far apart (i.e. larger central barrier width), the interaction between localized states of the well is very small and the wells behave as two independent identical single quantum wells. When the wells are brought closer by decreasing the central barrier, the energy levels interact, with one being forced to higher energies and the other to lower energies. This is analogous to the bonding and anti-bonding orbitals when two hydrogen atoms come together to form an ionized hydrogen molecule, H_2^+. In such a case the wave functions are, thus, $\Psi_{bonding} = \frac{1}{\sqrt{2}}(\psi_1 + \psi_2)$ and $\Psi_{anti-bonding} = \frac{1}{\sqrt{2}}(\psi_1 - \psi_2)$ for the ground and first excited states respectively [11]. The symmetric (n = 1) and anti-symmetric (n = 2) wavefunctions of the double QW with an arbitrary central barrier width are displayed in Figure 3.7 below, the former being lower energy.

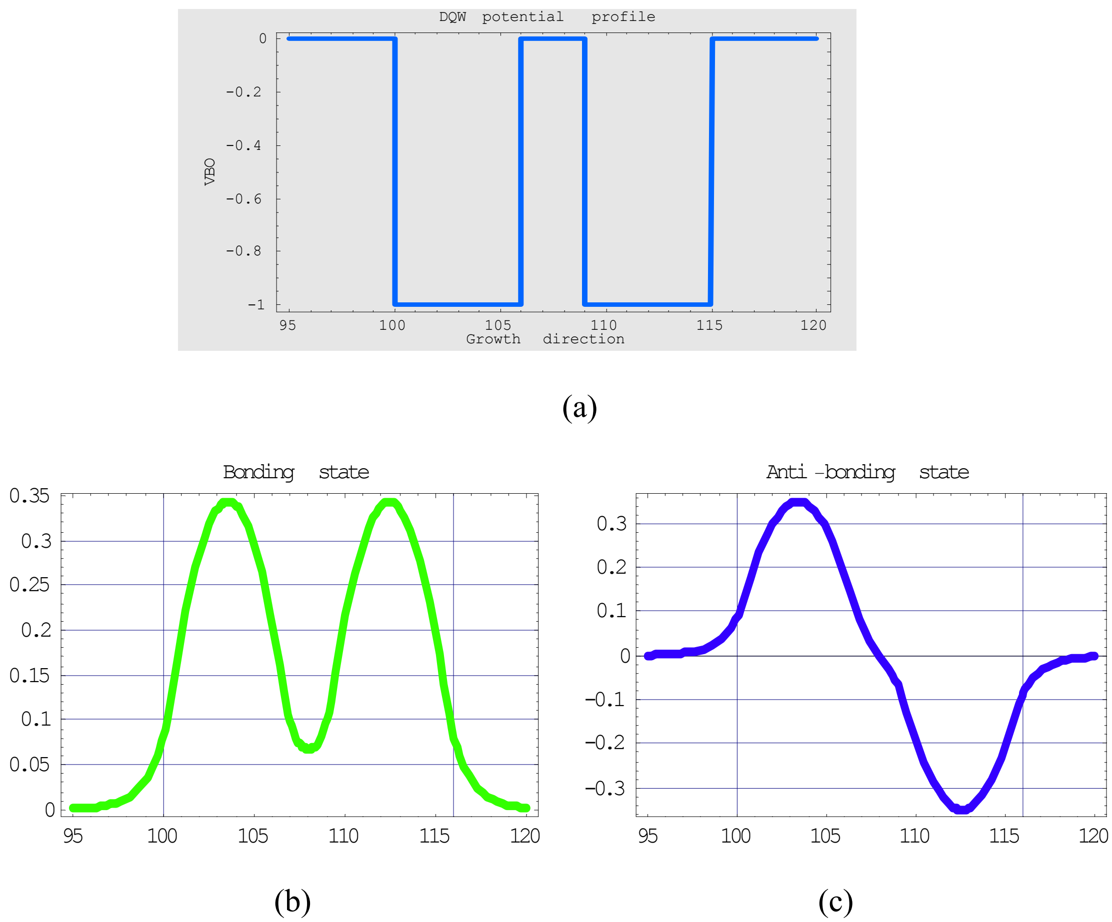


Figure 3.7 Symmetric double QW (a) potential profile (b) first confined state wave function (c) second confined state wave function

The electron exhibits the tunneling effect across the barrier. This feature is evident from the finite probability in the central barrier region in Figure 3.7 (b). The electron can not remain confined to one of the wells because of the tunnel effect. The tunnel effect lifts the degeneracy which is obvious from the symmetry of the problem, and this is accomplished by the symmetrical splitting of E_n as [35]:

$$E_n^{(Anti-bonding)} = E_n + \frac{1}{2}\delta E_n \quad \text{and} \quad E_n^{(Bonding)} = E_n - \frac{1}{2}\delta E_n$$

where the difference between the two levels is given by $\delta E_n = E_o e^{-R/L}$; $E_o = \dfrac{\hbar^2}{2mL^2}$, where R and L are inter-well distance and well width (characteristic length) respectively [35]. A numerical study of δE_n is given in Figure 3.8 which shows that the $E_n^{(Anti-bonding)} - E_n^{(Bonding)}$ decreases monotonically as the central barrier width increases and eventually degenerates.

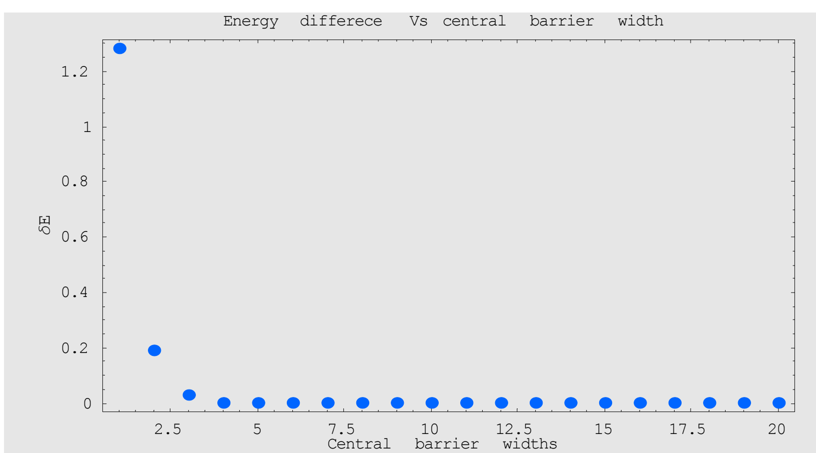
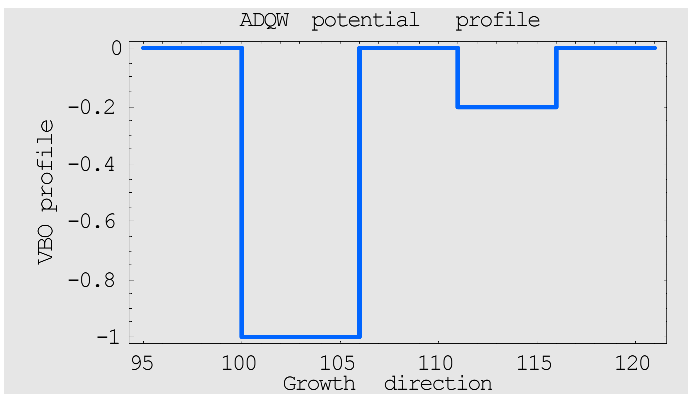


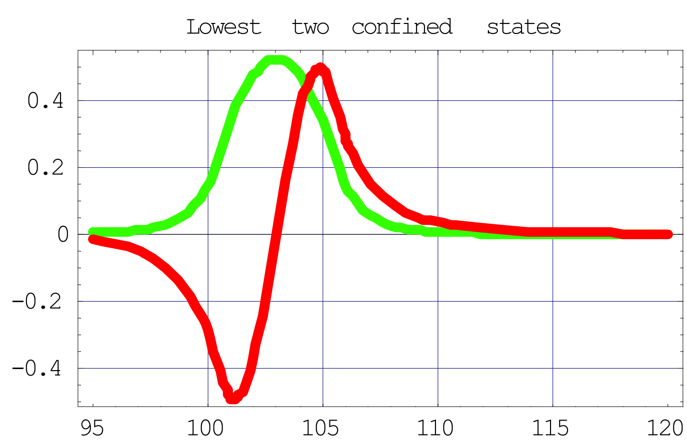
Figure 3.8 $E_n^{(A)} - E_n^{(B)}$ as a function of central barrier width of a DQW.

Parameters: n = 100, m = 7, γ = -1.0, u = central barrier width

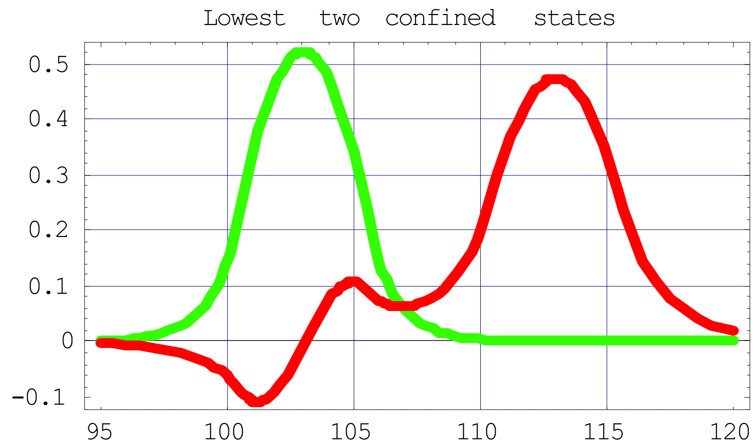
It is important to note that a typical asymmetric double quantum well (ADQW) may be formed by varying the well depths or widths. The asymmetry of DQW breaks the parity of wavefunctions and the localization of the electronic states can be controlled by changing width and depth of the wells. Figures 3.9 and 3.10 display the results of such calculations. As an example, the confined states for different depths of second well are given in Figures 3.9(b), (c) and (d). They show that the second confined state wavefunction gradually tends to localize in the second well as its depth increases. It is noticeable that the nodal structure is preserved.



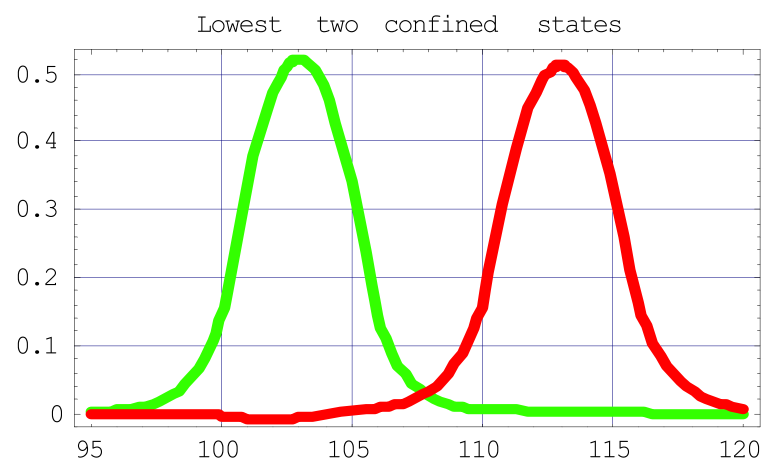
(a)



(b) second_VBO = 0.2*first_VBO



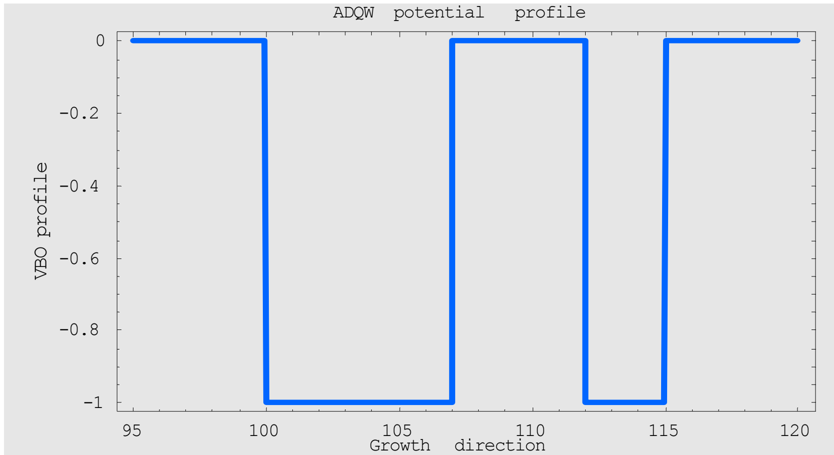
(c) second_VBO = 0.5*first_VBO



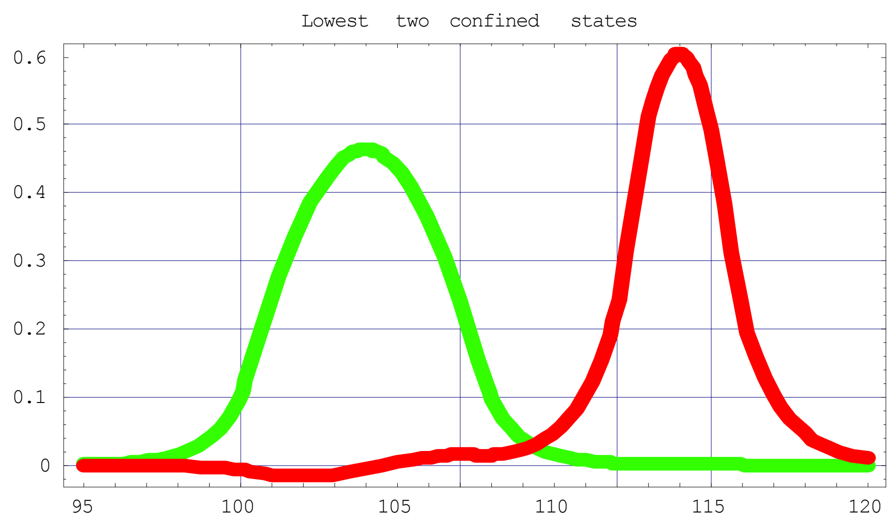
(d) second_VBO = 0.8*first_VBO

Figure 3.9 Asymmetry by well depths: ADQW (a) potential profile, and (b), (c) & (d) first confined state (Green) and second confined state (Red); n = 100, m = 5, u = 5, γ = -1.0 and first_VBO = -1.0 (see introduction for definition of notations)

The asymmetry of DQW may also be made by choosing different well widths and keeping well depths constant as shown in the Figure 3.10(a) below. However, it preserves similar properties as in the previous ADQW.



(a)



(b)

Figure 3.10 Asymmetry by well widths: ADQW (a) potential profile (b) first confined state (green) and second confined state (red); n = 100, fm = 7, sm = 3, u = 5, γ = -1.0 and VBO = -1.0

The asymmetric double QW results may be useful to describe the electron states in the systems HCl, NaCl, and others.

3.3 Quantum well with periodic boundary condition (i.e. infinite superlattice)

When periodic boundary conditions are applied to the QW problem, we obtain an infinite superlattice (SL). When many identical quantum wells interact the electrons and holes see a periodic potential, which can appear to be infinite in exactly the same way as bulk crystals. When this occurs, the electron and hole wave functions are no longer localized but are of infinite extent and equally likely to be in any of the quantum wells. They are said to occupy Bloch states [11].

A superlattice can have minibands; likewise, single QW has confined states. The Figure 3.11 shows the numerical energy solutions of a hypothetical SL solved by our simple TB scheme. The lowest energy miniband looks almost flat. Other higher energy minibands have the same periodicity as the first, although miniband minimum energy occurs at the edge of the SL Brillouin zone and at the center alternately. Optical transitions between minibands are exploited in the inter-miniband laser, which is a potential high-power source of mid-infrared radiation [11]. It is important to note that the periodicity and the strength of interaction between the adjacent wells in SL are under our control, so we may make devices with a tailor-made band gap.

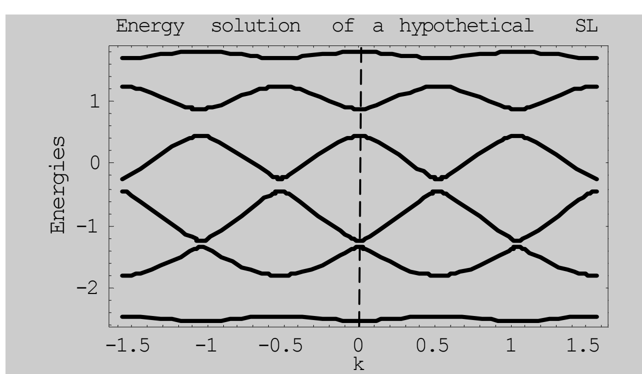


Figure 3.11 Energy solutions of a hypothetical superlattice

One of the most remarkable quantum phenomena in a miniband of superlattice is the occurrence of 'Bloch oscillations.' There exists an ideal oscillatory motion of an electron in a miniband of superlattice with sufficiently larger lattice parameter and relaxation time. The electron in such miniband oscillates within the same band with frequency $\omega_B \approx 10^{13}$ rad/sec, provided electric field of 10^4 Volt/cm and lattice constant of GaAs well of 100 $\overset{o}{A}$ [20, Appendix E].

3.4 Finite superlattices and multiple quantum well

An infinite superlattice is an ideal structure which actually does not exist. Real layered structures contain a finite number of quantum wells. As it can be seen in a numerical result (Figure 3.12), the lowest confined state energy maintains a constant value after a certain number of wells (above number of quantum wells = 15 in this example); this means the system resembles an infinite superlattice when there is sufficient number of wells.

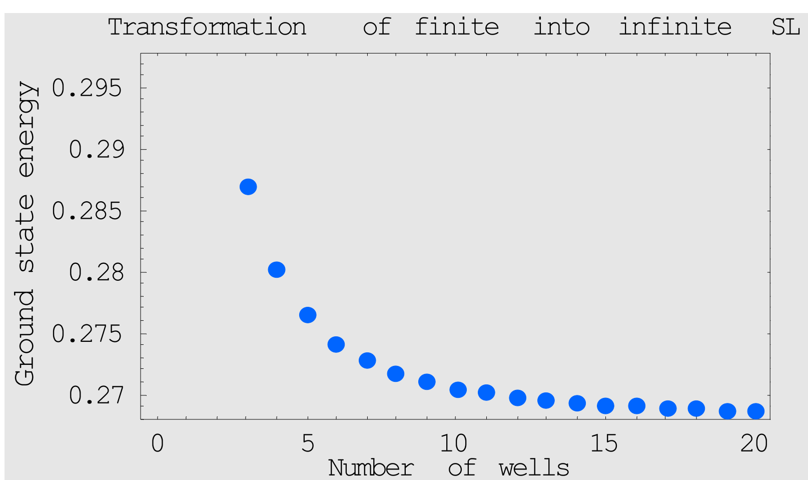


Figure 3.12 The ground state energy versus number of quantum wells in SL

Figure 3.13 shows the lowest confined state wave function of a finite superlattice as the wave function associated with each well has a significant overlap with that of the adjacent well. Also, because of the finiteness of the structure, the electron is not equally likely to be in any of the wells. On the contrary, the first confined state wave function in Figure 3.14 (b) (compared with Figure 3.13) reaches zero between the wells which describes the system being clearly a MQW [11].

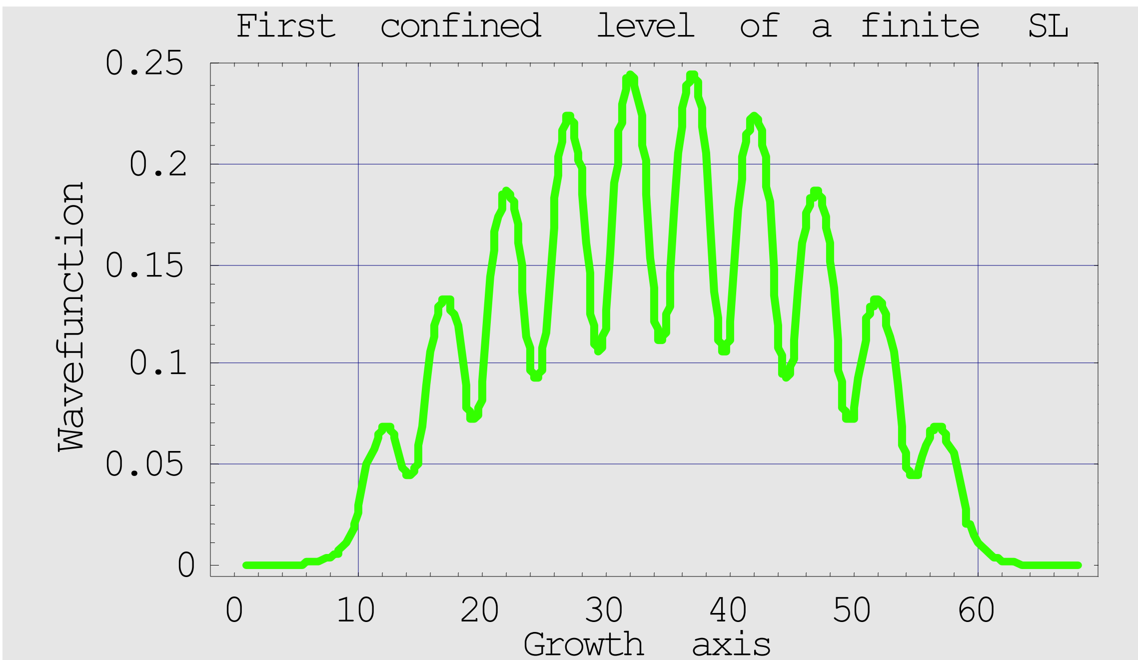
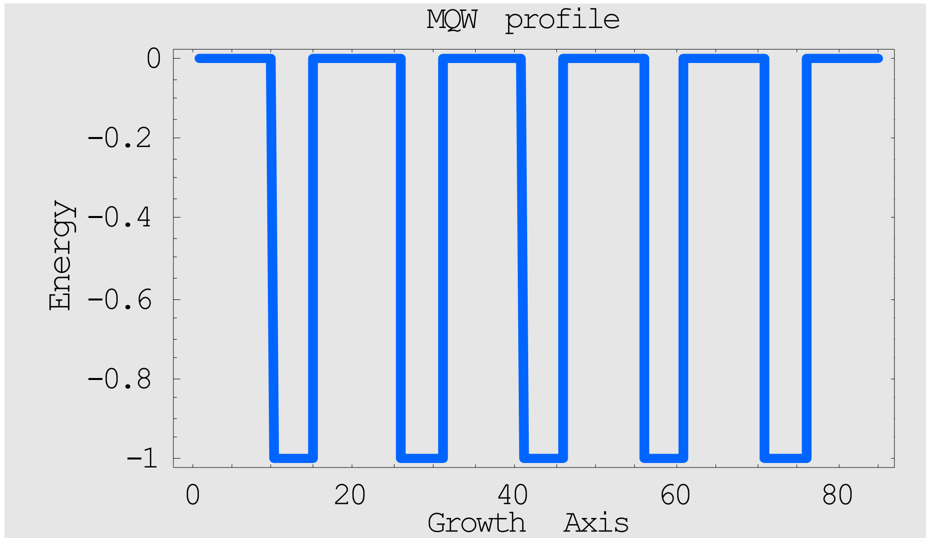
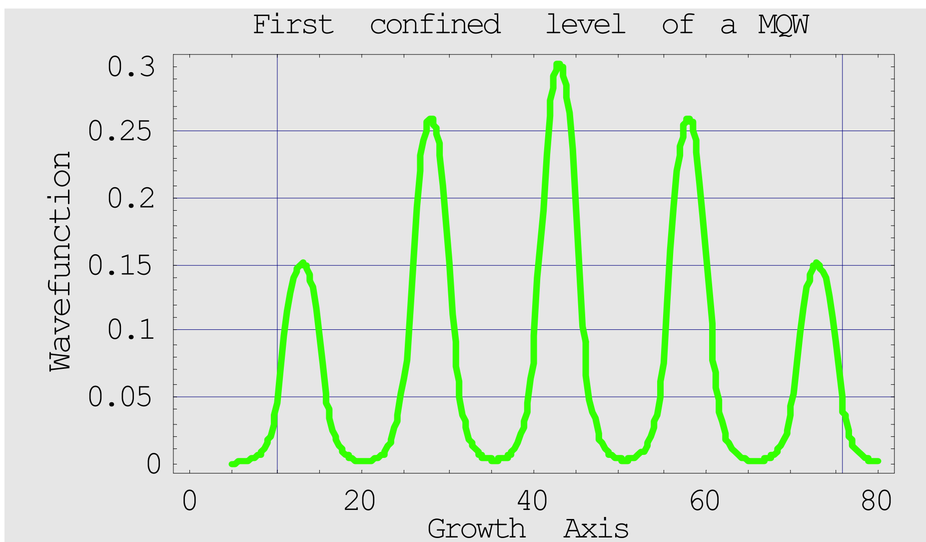


Figure 3.13 The ground state wave function of a finite SL: n = 10, m = 3, u = 2, γ = -1.0, VBO = -1.0



(a)



(b)

Figure 3.14 (a) MQW potential profile (b) The ground state wave function of a MQW: $n = 10$, $m = 5$, $u = 10$, $\gamma = -1.0$, VBO = -1.0 (see introduction for definition of notations)

MQW structures are used because of large exitonic effects, which persist up to room temperature. The uses of these devices are, therefore, often related to optical properties [9, 19].

3.5 The parabolic quantum well

The parabolic QW is analogous to the harmonic oscillator for which the potential is defined as $V(z) = \frac{1}{2}kz^2$. We used a truncated parabola that is schematically shown in Figure 3.15.

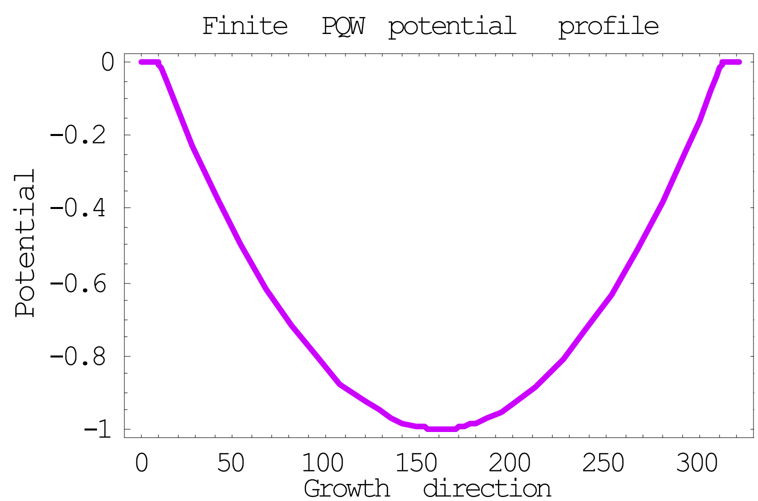


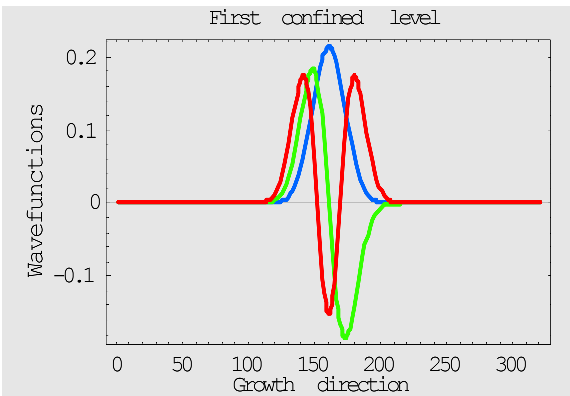
Figure 3.15 Parabolic quantum well profile: n = 10, m = 301, γ = -1.0, VBO = -1.0

The parabolic quantum well is a good testing ground for the accuracy of the numerical method introduced. The energy levels are quantized in the parabolic QW and should be $E_n = E_0 + \hbar\omega n$. The lowest state or zero point energy is E_0. Above the zero point energy the equal-spaced energy steps form a ladder which has prompted suggestions for exploitation in non-linear optics [11]. The accuracy test of the numerical solution for a finite parabolic QW formed with parameters (barrier = 10, well = 301, VBO = -1.0, γ = -1.0) is given in the table 3.2. The energy step $\hbar\omega$ between the levels is found to be constant as expected. However, deviation occurs at high energy.

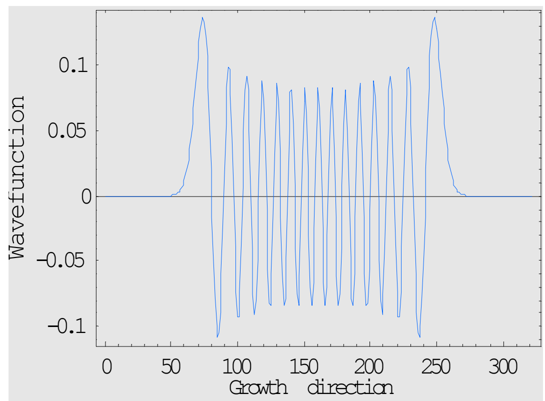
Table 3.2 Accuracy tests and $\hbar\omega$ for lowest 20 confined states of the parabolic QW

Nos.	E(n)	E(n))/[2E(1)]	(E(n+1)-E(n))/[2E(1)] ~1	$\hbar\omega$
1	0.00661977	0.5000	0.999585	0.0132341
2	0.01985380	1.49959	0.998756	0.0132231
3	0.03307690	2.49834	0.997925	0.0132121
4	0.04628900	3.49627	0.997093	0.0132011
5	0.05949000	4.49336	0.996260	0.0131900
6	0.07268010	5.48962	0.995426	0.0131790
7	0.08585910	6.48505	0.994592	0.0131679
8	0.09902700	7.47964	0.993756	0.0131569
9	0.11218400	8.47339	0.992918	0.0131458
10	0.12533000	9.46631	0.992080	0.0131347
11	0.13846400	10.4584	0.991241	0.0131236
12	0.15158800	11.4496	0.990401	0.0131125
13	0.16470000	12.4400	0.989560	0.0131013
14	0.17780200	13.4296	0.988717	0.0130902
15	0.190892	14.4183	0.987874	0.0130790
16	0.203971	15.4062	0.987029	0.0130678
17	0.217039	16.3932	0.986183	0.0130566
18	0.230095	17.3794	0.985336	0.0130454
19	0.243141	18.3647	0.984488	0.0130342
20	0.256175	19.3492	0.983639	0.0130229

The Figures 3.16(a) and 3.17(a) display the wavefunctions and eigenenergies of the first three confined states of the more realistic finite parabolic QW considered in the numerical method respectively. The wavefunctions resemble the expected Hermite polynomial; and the energy levels are equally spaced as expected.



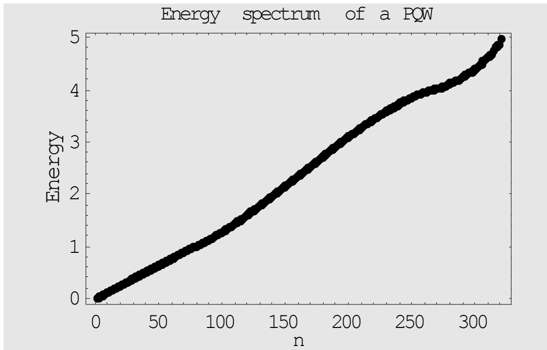
(a)



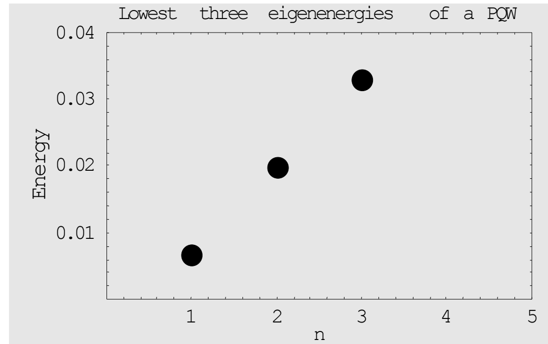
(b)

Figure 3.16 A finite parabolic QW (a) the lowest three: Blue (n =1), Green (n =2), Red (n =3) and (b) 29th confined eigenstates

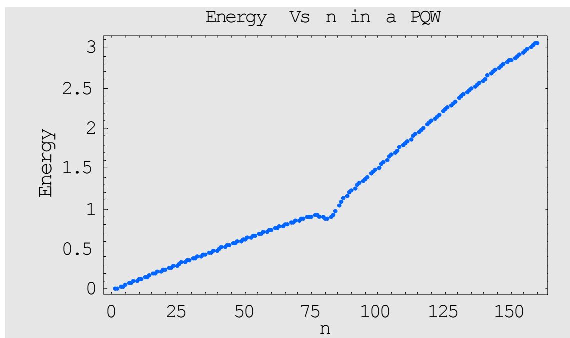
It can be noted from the Figure 3.16(b) that at higher quantum numbers the average behavior of rapidly oscillating quantum wavefunctions (or the probability of finding the particles) is correspondent to classical limit as defined by $(\xi_0^2 - \xi^2)^{-1/2}$, where $\xi = \alpha z$ and ξ_0 is the amplitude of the oscillator with the given energy [25].



(a)



(b)



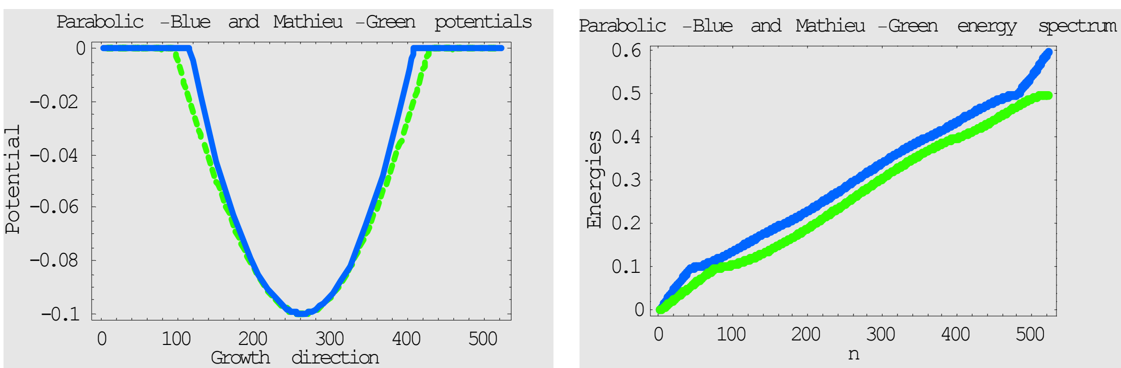
(c)

Figure 3.17 Eigenenergies of (a) all confined and unconfined states (b) first three confined states of finite parabolic quantum well (PQW) and (c) $E = \beta n$ plot. Numerical parameters: n = 10, m = 301, γ = -1.0, VBO = -1.0. The dip in the graph may be due to the VBO limit

The deviation of eigenenergies at higher n in the parabolic quantum well is due to the finite height of the potential barrier. We also plotted $E = \beta n$, where $\beta = \hbar\omega = E(n) - E(n-1)$, as shown in Figure 3.17(c). We found a straight line for up-to a certain point within the PQW. Beyond the point the energy is not equally spaced as they are unconfined.

3.6 Mathieu potential well

Some of the interesting features of a parabolic potential may be related also in a smooth periodic superlattice whose potential profile follows a $V(z) = A\cos(z)$ law, the Mathieu potential [39, Appendix D]. These systems may be solved analytically [20], and experimentally they are created by controlling the alloy composition. The potential profile of one period of Mathieu potential compared with parabolic potential is given in Figure 3.18(a).



(a) (b)

Figure 3.18 Mathieu versus Parabolic (a) potential profile and (b) energy spectrum.

Parameters: n = 10, m = 501, γ = -0.1, VBO = -0.1

The Figure 3.18(b) and table 3.3 shows the calculation of difference between Mathieu and parabolic QW energies. At lower values of n the eigenenergies tend to be closer; however, at higher values of n the difference between Mathieu and parabolic energies is larger. Parameters for table 3.3: n = 10, m = 501, γ = -0.1, VBO = -0.1 (see introduction for definition of notations)

Table 3.3 Difference between Mathieu and parabolic eigenenergies

Nos.	Mathieu (M)	Parabolic (P)	(M) – (P)
1	0.00179921	0.000682691	0.00111652
2	0.00434074	0.002047490	0.00229326
3	0.00689403	0.003411120	0.00348292
4	0.00944684	0.004773580	0.00467326
5	0.01199450	0.006134860	0.00585964
6	0.01453470	0.007494980	0.00703972
7	0.01706610	0.008853920	0.00821215
8	0.01958770	0.010211700	0.00937604
9	0.02209910	0.011568300	0.01053080
10	0.02459970	0.012923700	0.01167600
11	0.0270892	0.014277900	0.01281130
12	0.02956740	0.015631000	0.01393650
13	0.03203410	0.016982800	0.01505130
14	0.03448910	0.018333500	0.01615560
15	0.03693230	0.019683000	0.01724930
16	0.03936360	0.021031300	0.01833230
17	0.04178290	0.022378400	0.01840450
18	0.04419000	0.023724300	0.02046570
19	0.04658500	0.025069000	0.0215160
20	0.04896780	0.026412600	0.02255520

In order to analyze the properties of the Mathieu potential as periodic counterpart of the harmonic oscillator we first studied a single cosine well (as in table 3.3) and then introduced the periodic boundary condition. The band structure of an ideal Mathieu SL is given in Figure 3.19, the numerical solution matches the analytical band structure [20].

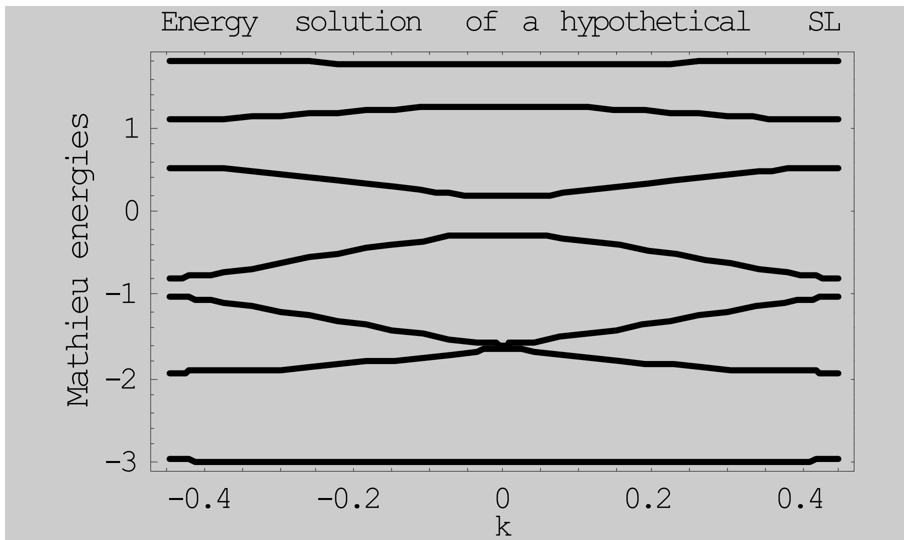


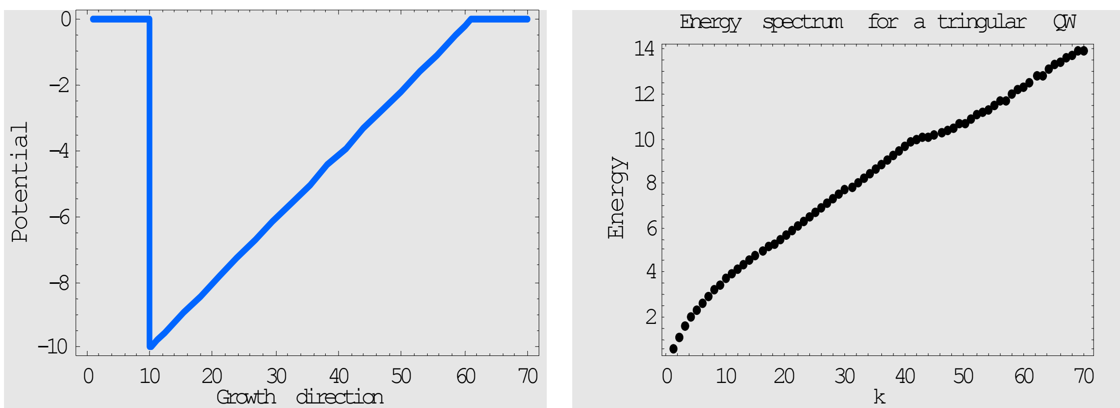
Figure 3.19 Energy bands for the periodic Mathieu potential. Parameters: n = 3, m = 4, γ = -1.0, VBO = -1.0

3.7 Triangular quantum well (TQW) and cascade laser

An electron in an electric field obeys the 1-D Hamiltonian, $\frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial z^2} + e\mathcal{E}z = H$,

which provides energies that extend from $-\infty$ to $+\infty$ in a continuous way. The quantum mechanical solution of this problem in a crystal was first derived by Wannier in the '60s and is usually referred as Wannier-Stark quantization (Appendix E). The energy spectrum is a set of levels that are all equally spaced of an amount $\Delta E = e\mathcal{E}a$; the wavefunctions are spatially localized and are described by Airy functions. The TB scheme provides straight forward way to study these effects in the case when boundary conditions are applied. We studied these in a triangular well (saw tooth potential) without and with PBC.

The single triangular well Hamiltonian does not commute with the spatial inversion and, as expected, the wave functions are not symmetric and resemble the Airy functions (at least at low energies). The energy levels are equally spaced as in the case of the harmonic oscillator [39]; however, deviations occur for large n. The potential profile and the electron energy spectrum for TQW are shown in Figure 3.20.



(a) (b)

Figure 3.20 (a) TQW potential profile, and (b) energy specturm. Parameters: n = 10, m = 50, γ = -1.0, VBO = -10.0 (see introduction for definition of notations)

Some numerically calculated wave functions are displayed in the Figure 3.21 for the above TQW profile.

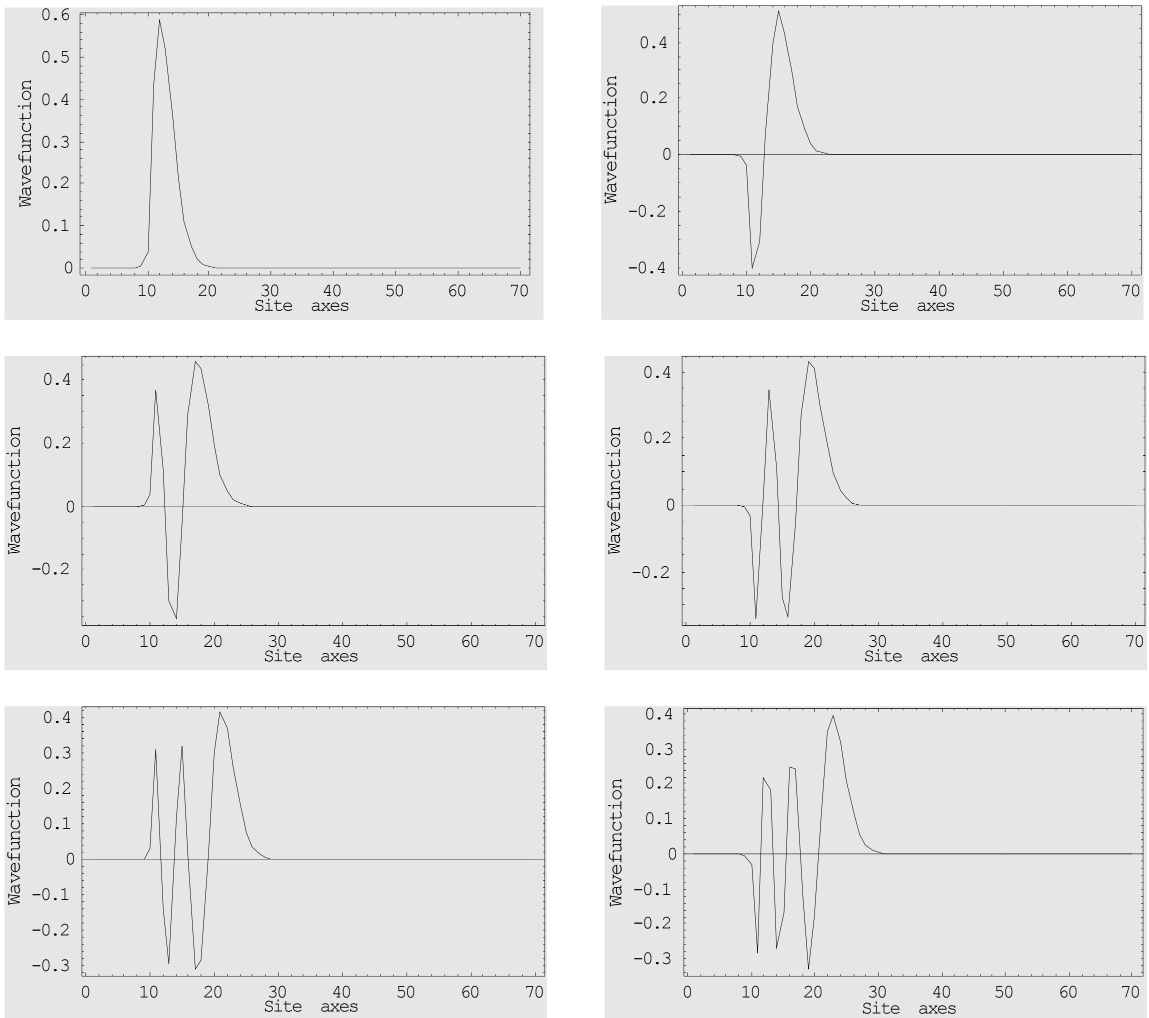


Figure 3.21 Wavefunctions for lowest six states for a TQW

The TQW may be modified and refined into other forms of quantum structures such as trapezoidal, V-shaped etc. A change is apparent in wavefunctions as given in Figures 3.21 and 3.22: fourth confined state wavefunctions for the triangular and V-shaped quantum wells.

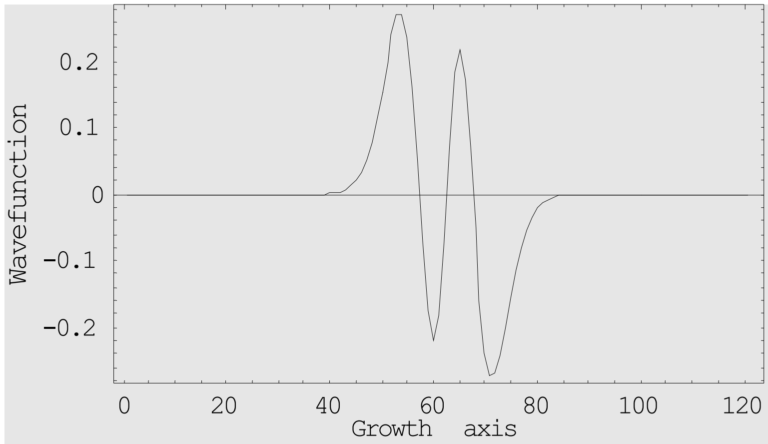


Figure 3.22 Numerical wavefunction (fourth confined state) in a V-shaped quantum well.

Parameters: n = 10, m = 101, γ = -1.0, VBO = -5.0

The purpose of seeking diversity of such shapes is to create potential new quantum states and exploit their wavefunctions, energy and momentum for novel electronic and optical properties which may be exploited for new device concepts.

The introduction of well-well interaction removes the degeneracies similar to the case of square QW and SL. An extremely interesting case for the applications is an electric field superimposed to a SL. A profile of quantum cascade structure [19] formed in this way is shown in Figure 3.23 below. This potential profile is used sometimes for quantum cascade photomultipliers. The energy spectrum consists of equal space ladder-like steps from well to well of cascade and the wavefunction are localized in each well although the tunneling is not negligible.

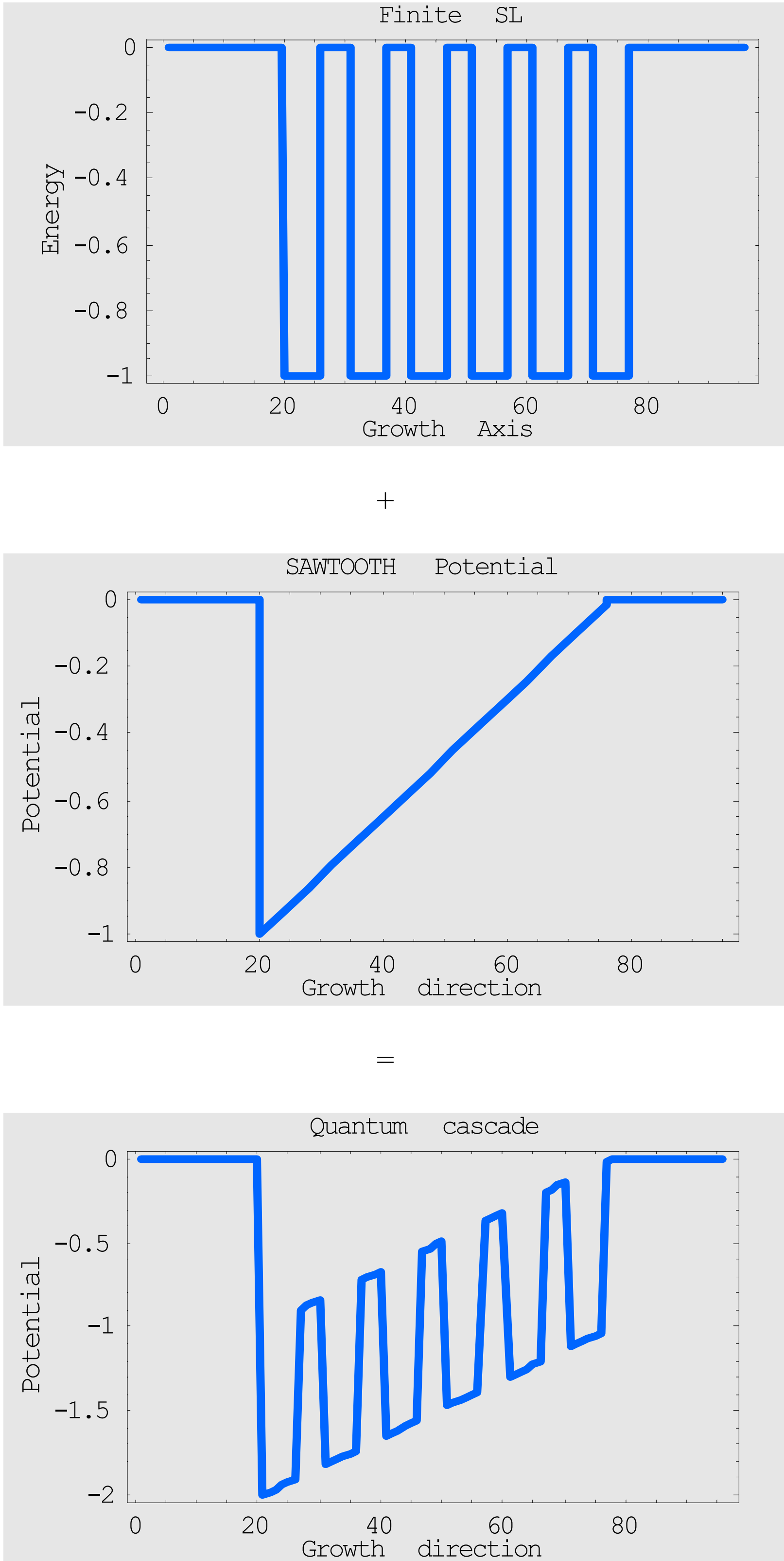
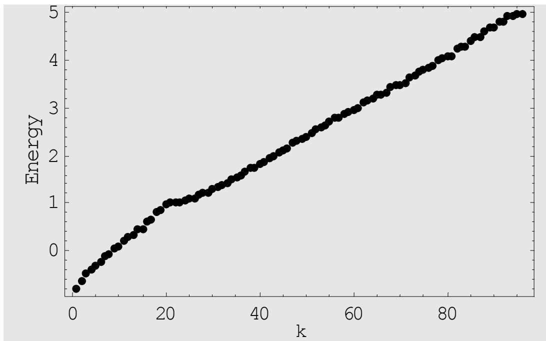
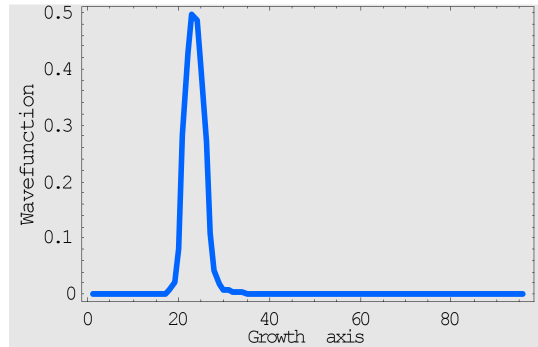


Figure 3.23 Quantum cascade structure

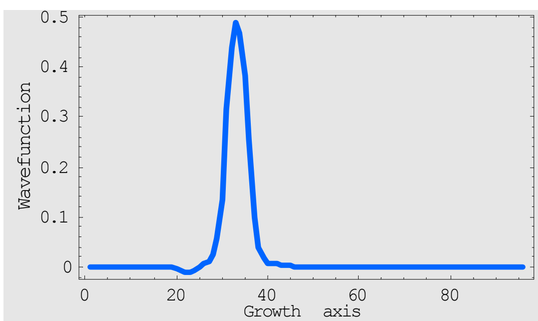
The calculation of energy spectrum and first few wavefunctions of a quantum cascade are shown in Figure 3.24.



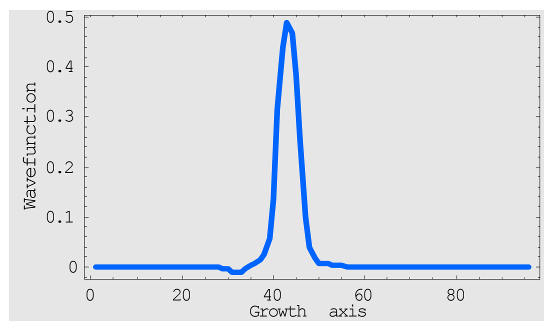
(a) energy spectrum



(b) first confined state



(c) second confined state



d) third confined state

Figure 3.24 Quantum cascade (a) energy spectrum (b), (c) & (d) lowest three wavefunctions. Parameters: n = 20, m = 6, u = 4, γ = -1.0, VBO = -1.0. The kink in the energy spectrum is due to the VBO limit

CHAPTER IV

SELF-CONSISTENT FIELD METHOD

The TB scheme used so far does not allow charge rearrangement and the profile used to describe the potential is fixed *a priori*. In many situations (e.g. modulation doping), however, the potential is influenced by charge localization in the nanostructures. Our strategy to refine the initial potential profile makes use of self-consistency to determine iteratively the corrections to the potential. The self-consistent cycle is illustrated in the flow chart (Figure 4.1) below [40].

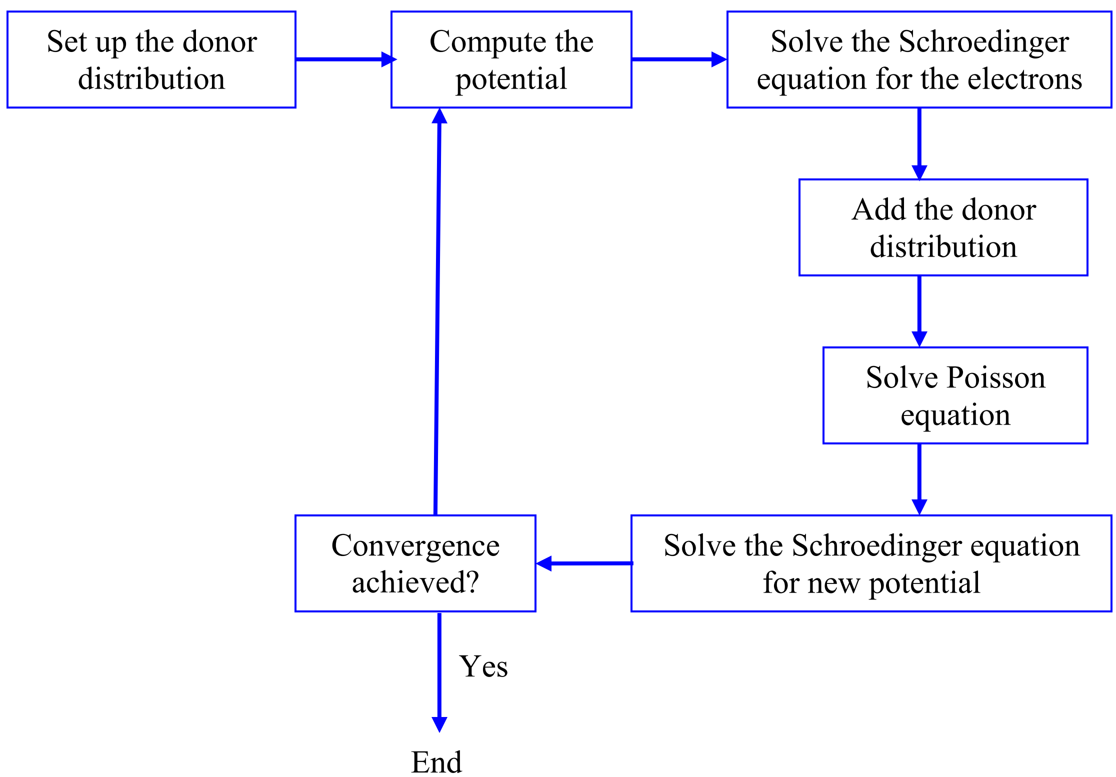


Figure 4.1 Self-consistent method cycle

At the beginning of the self-consistency process, a trial tri-diagonal Hamiltonian matrix is constructed with the approximate potential profile (e.g. band edge potential + potential created by the donors). The diagonalization of the Hamiltonian matrix produces a set of eigenvectors. These eigenvectors and an occupation function are used to calculate the electron densities at space coordinates. The electron density, $\rho(z)$, is the probability of finding an electron at z. Mathematically, this is defined by the sum of orbital densities,

$$\rho(z) = 2\sum_{i}^{n}\psi_i^2(z), \tag{4.1}$$

where 2 is spin degeneracy factor, i.e. 2 electrons in each n. Having calculated the electron densities, they are fed into a Poisson's solver which solves

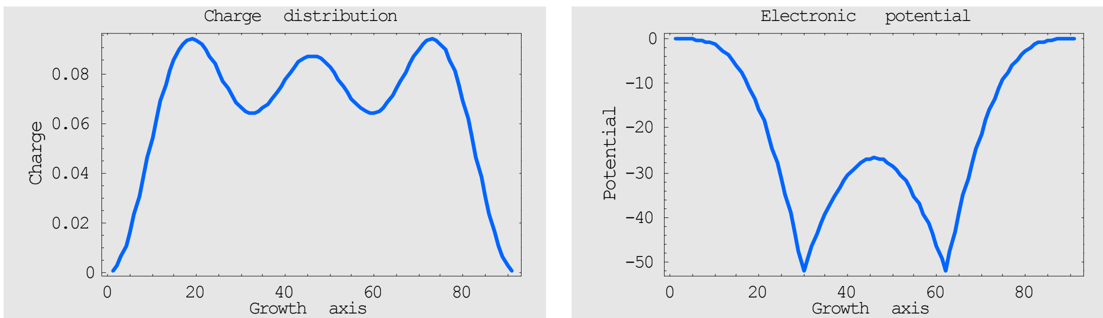
$$\frac{d^2V_\rho(z)}{dz^2} = - \frac{4\pi\rho(z)}{\varepsilon} \tag{4.2}$$

for the potential $V_\rho(z)$ at all space coordinates [40].

The parameters of the SCF calculation include the number of electrons and the charge distribution of the donors (static positive charge). First we analyze the case of double δ – doping (inside a particle in a box (PIB) problem) in which two n-doped mono-layers are separated by a neutral layer. Then we describe the potential profile for a doped heterostructrue. Of course, neutrality is imposed on all the systems.

4.1 δ – doping

Two doped layers are introduced in the PIB problem creating an initial charge distribution and potential as shown in Figure 4.2. The interaction between these layers is noticeable from the charge distribution and potential profile.

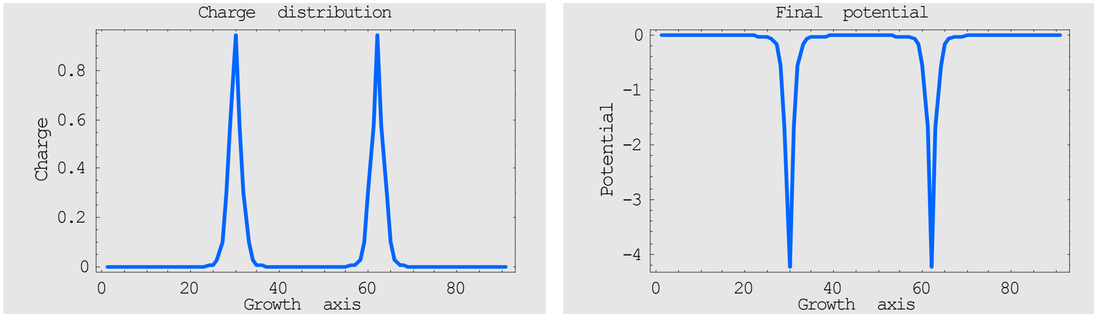


(a) (b)

Figure 4.2 Smaller ε case: Initial (a) charge distribution, and (b) potential.

Parameters: n = 30, m = 31, γ = -0.8, ε = 10, # of electrons = 12 (L = 6)

The dielectric constant ε is chosen initially equal to 10. The self consistent cycle rearranges the charge so that the doped layers are fully screened (i.e. there is no interaction between the doped layers due to self-consistent effect) as expected by simple electrostatics. The screening length for the parameters used is about 5 layers and the two "impurities" do not interact. Convergence is achieved in about 90 iterations. The final charge distribution and potential are shown in Figure 4.3.

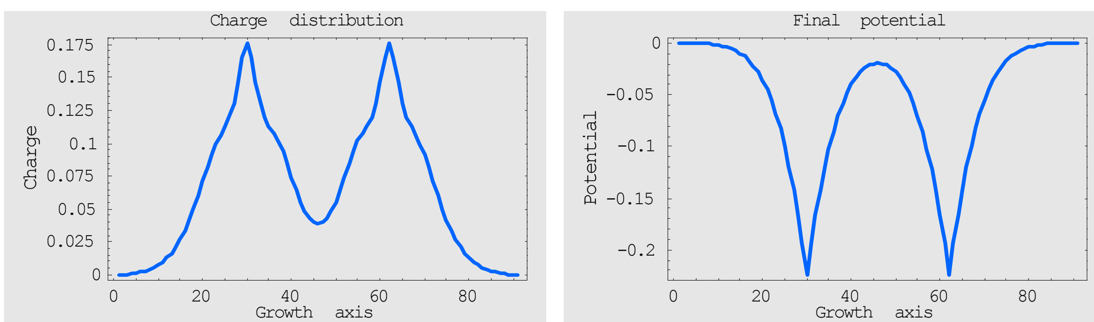


(a) (b)

Figure 4.3 Smaller ε case: Final (a) charge distribution, and (b) potential.

Parameters: n = 30, m = 31, γ = -0.8, ε = 10, # of electrons = 12 (L = 6)

The increase in the dielectric constant results in smoother potential. At large ε the dopants are less active and the electrons do not localize as strongly around the δ – layers. This is shown in Figure 4.4.

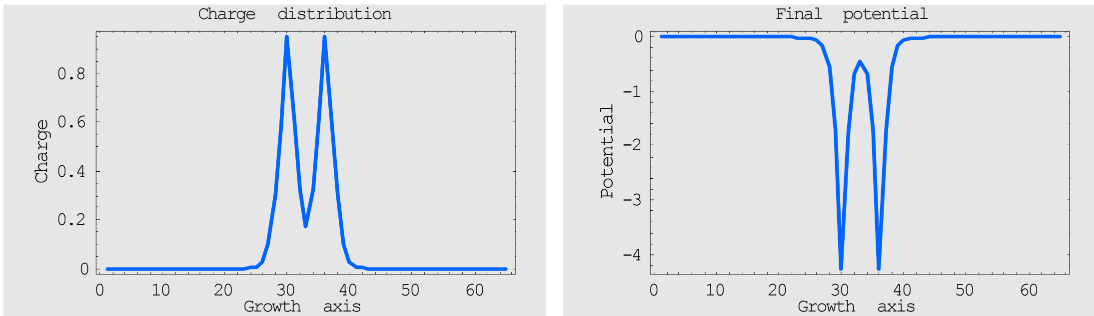


(a) (b)

Figure 4.4 Larger ε case: Final (a) charge distribution, and (b) potential.

Parameters: as in Figure 4.2, but ε = 1000

Interaction between doped layers may be achieved by decreasing the layer-layer distance to few (say 5) layers. The final charge distribution and potential are in Figure 4.5 for a typical few layer case.

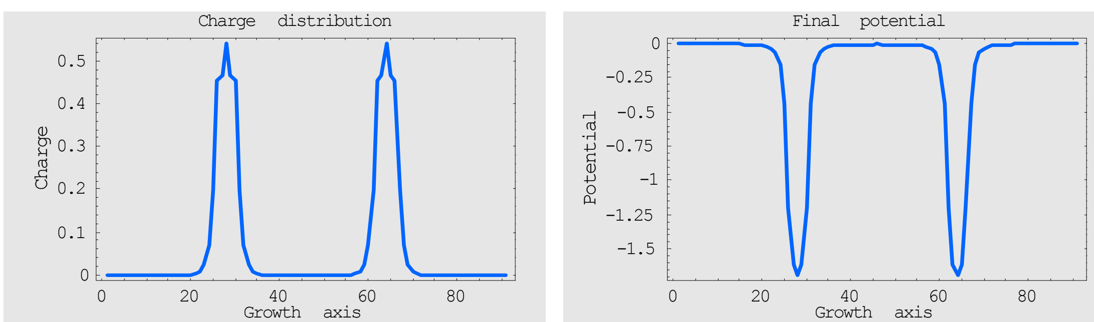


(a) (b)

Figure 4.5 Layer-layer distance smaller: Final (a) charge distribution, and (b) potential.

Parameters: n = 30, m = 5, γ = -0.8, ε = 10, # of electrons = 12 (L = 6)

Perfect δ – doping is not easy to achieve experimentally, usually more than one layer is doped in the growth. This results in a broader electron charge distributions as shown in Figure 4.6.



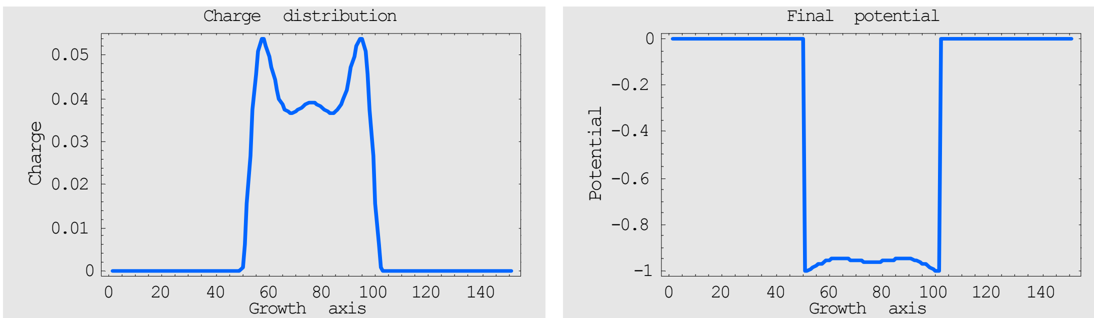
(a) (b)

Figure 4.6 Broader donor distribution: Final (a) charge distribution, and (b) potential.

Parameters: n = 30, m = 31, γ = -0.8, ε = 10, # of electrons = 12 (L = 6)

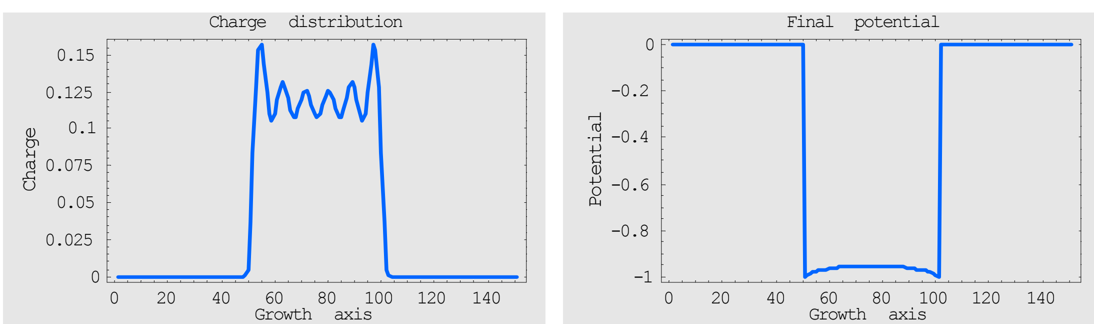
4.2 Doped heterostructures

In a doped QW the abrupt step-like potential changes slightly creating a double well system that tends to localize the charge at the interfaces. This is explored by calculating charge density with different number of electrons, 4 (L= 2) and 12 (L=6) electrons, as shown in Figures 4.7 and 4.8 respectively.



(a) (b)

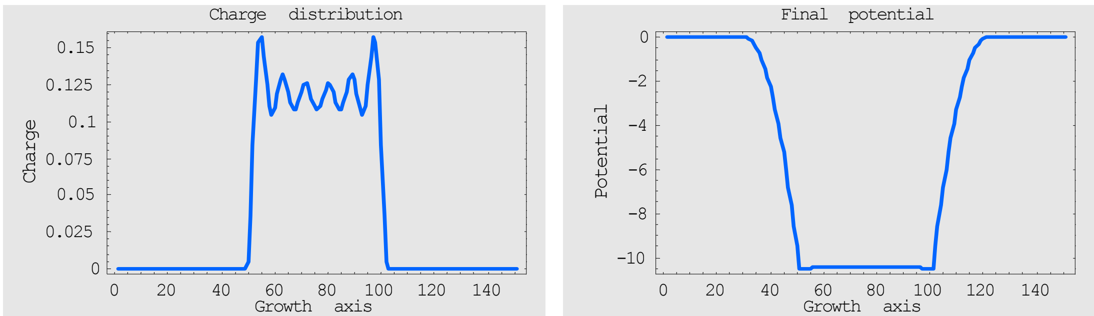
Figure 4.7 L smaller case: Final (a) charge distribution, and (b) potential. Parameters: n = 50, m = 51, VBO = 1.0, γ = -0.8, ε = 10



(a) (b)

Figure 4.8 L larger case: Final (a) charge distribution, and (b) potential. Parameters: n = 50, m = 51, VBO = 1.0, γ = -0.8, ε = 10

The charge distribution and potential for a uniform donor distribution for QW are shown in Figure 4.9. The potential is now obtained smoother than the previous cases. The smoother potential is more favorable to nature.



(a) (b)

Figure 4.9 Broader donor distribution: Final (a) charge distribution, and (b) potential.

Parameters: n = 50, m = 51, VBO = 1.0, γ = -0.8, ε = 10, # of electrons = 12 (L = 6)

CHAPTER V

SUMMARY AND CONCLUSIONS

In this project I have qualitatively described the basic features of the electronic properties of semiconductor microstructures, particularly quantum well, superlattice and multiple quantum well systems using tight binding calculations. Heterostructures are formed from layering different semiconductors or combinations of semiconductors with metals or oxides [3]. They are usually fabricated by epitaxial growth techniques and are applied in a host of devices.

The phenomenon of quantum confinement effect is studied in several systems that are meaningful from an applications point of view. It has been shown that simple tight-binding approaches capture the electronic structure phenomena in many different situations. The method provides a conceptually transparent approach to device physics that may go beyond approximations such as effective mass and envelope function.

We analyzed square well potentials, matching results obtained with the effective mass approximations. We also analyzed the symmetric and asymmetric square well potential, reproducing the formation of bonding and anti-bonding levels as in H_2. Multi-quantum-wells and superlattices were studied to investigate the effect of periodicity. In all these calculations we pointed out the limitation of the simplified approach we used.

We also studied parabolic quantum wells to mimic the properties of the harmonic oscillator. Triangular quantum wells were analyzed because of their importance when electric fields are applied to crystals. Last but not least we investigated effects induced by the charge rearrangement in a QW by self-consistent field method.

The main goal of the present work was to apply a simple TB computational technique to a variety of quantum mechanical problems that are of technological significance for present and future applications in electronic and opto-electronic devices. As my work demonstrates, simple TB models are pedagogically sound and capture the core of many quantum effects.

APPENDICES

Appendix A

List of some Mathematica notebooks constructed and used in the computational analysis

1. one_dim_chain
2. one_dim_chain_pbc
3. inf_sq_well_analytical
4. inf_sq_well_energy_vs_well_widths
5. QW_single_isolated
6. QW_single_isolated
7. QW_single_energy_vs_well_width
8. QW_single_energy_vs_well_depth
9. QW_pbc
10. QW_scf
11. HO_analytical
12. HO_numerical
13. HO_pbc
14. mathieu1_pot
15. mathieu3_pot
16. mathieu3_pot_pbc
17. triangular_pot_b_slope
18. triangular_pot_V_shape
19. triangular_trapezoidal_pot

20. triangular_pot_pbc
21. double_QW
22. double_QW_E_vs_barr_widths
23. double_QW_Asym_well_widths
24. SL_MQW
25. SL_MQW_E1_vs-numb_of_well
26. cascade
27. scf_delta
28. scf_QW

Appendix B

Notebook syntax for quantum well

File name: QW_single_isolated

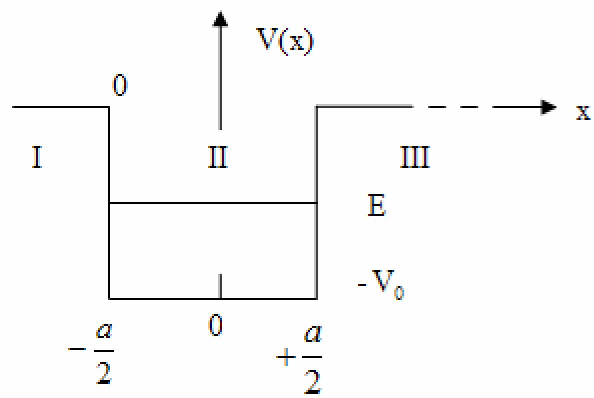
```
(*This program calculates Eigenvalues and Eigenvectors in an
Isolated Quantum Well in 1-Dimension*)
(*-------------------------*)
(*--inputs==============================================*)
n=100;          (*-atoms in the barriers--*)
m=10;         (*-atoms in the well: width of well--*)
e=0.0;        (*-on-site energy reference--*)
t=-1.0;       (*--NN interaction energy--*)
VBO=-2.0;(*-Valence Band Offset: depth of potential well-*)
  (*-------------------------*)
  (*-------------------------*)
  (*--Construction of Hamiltonian matrix ===============*)
  A=Table[e,{i,n}];
  B=Table[VBO,{i,m}];
  Diag=Join[A,B,A];
  Hopp=Table[t,{i,2n+m-1}];
  H=Table[Switch[r-c,1,Hopp[[r-1]],0,Diag[[r]],-1,Hopp[[c-
1]],_,0],{r,2n+m},{c,2n+m}];
             (*--r=row index of Hamiltonian matrix--*)
             (*--c=column index of Hamiltonian matrix--*)
  MatrixForm[H];
  (*------------------*)
  (*--Checking if the diagonal elements fit the POTENTIAL
PROFILE==============================================*)
  Print["POTENTIAL PROFILE"];
  P=Table[VBO];
  v[i_]:= Which[i<n,0,n≤i≤n+m+1,P,n+m+1<i,0];
  v1=Plot[v[i],{i,1,2n+m},Frame→True,FrameLabel→{"i-
Cordinate","V(i)-
Potential"},PlotRange→Automatic,PlotStyle→Hue[0.8],RotateLa
bel→True];
  (*------------------*)
  (*--calculation of eigenvalues ========================*)
  eval=Eigenvalues[H];
  P=Ordering[eval];
(*--the Figures in P indicate the relative positions of
eigenvalues--*)
  Sort[eval];
(*--this sorts eigenvalues out in increasing order--*)
  EPS=Sort[eval]-VBO-2t;  (
*--adjusting the energy reference--*)
```

```
  ListPlot[EPS,PlotRange→{0,8},PlotStyle→PointSize[0.02]];
(*--this is energy band of QW--*)
  (*------------------------*)
  (*--calculation of eigenvectors=====================*)
  evec=Eigenvectors[H];
  (*-----------*)
  (*--Multiple plot of eigenvectors-==================-*)
  (*----*)
  Interpolation[evec[[1]],InterpolationOrder→1];
  Interpolation[evec[[2]],InterpolationOrder→1];
  Interpolation[evec[[3]],InterpolationOrder→1];
  psi1=Plot[Evaluate[Interpolation[evec[[1]]]][x],{x,n-
5,n+m+5},PlotLabel->"First confined
level",PlotStyle→Hue[0.3],Frame→True,GridLines→{{n,n+m+1},A
utomatic}];
  psi2=Plot[Evaluate[Interpolation[evec[[2]]]][x],{x,n-
5,n+m+5},PlotLabel->"Second confined
level",PlotStyle→Hue[0.0],Frame→True,GridLines→{{n,n+m+1},A
utomatic}];
  psi3=Plot[Evaluate[Interpolation[evec[[3]]]][x],{x,n-
5,n+m+5},PlotLabel->"Third confined
level",PlotStyle→Hue[0.7],Frame→True,GridLines→{{n,n+m+1},A
utomatic}];
(*-"Show" will bring graphs psi1,psi2 and psi3 together--*)
  Show[{psi1,psi2,psi3},PlotLabel->"First three
wavefunctions together",GridLines→{{n,n+m+1},Automatic}];
  (*-----*)
  (*--eigenvector plots separately==================-*)
  L = 10;    (*-- number of states (energy levels) whose
wavefunctions to be observed -- *)
  Print["FOLLOWING ARE THE WAVEFUNCTIONS AT EACH LEVEL"];
  Do[Print["E(",k,")=",EPS[[k]]];
ListPlot[evec[[k]],PlotJoined→ True,PlotRange→{{n-
5,n+m+5},{-1,1}},GridLines→{{n,n+m+1},Automatic}],{k,1,L}]
  (*---------------*)
  (*--End program--*)
  (*--*)
  (*-============================================-*)
  (*------------------------*)
```

Appendix C

Finite Square Well Potential

The wave function involved in TISE for a finite square well potential reveals many interesting qualitative characteristics of quantum mechanical systems. The bound state (E< 0) finite square well potential as shown below



Finite square well potential

$$V(x) = \begin{cases} -V_o, & |x| \le \frac{a}{2} \\ 0, & |x| > \frac{a}{2} \end{cases}$$

and the corresponding Schroedinger's equation, $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$, are set in one-dimension. V(x) is symmetric about x = 0, ψ^2 represents the probability, and ψ (x) can be either an even or an odd function of x.

The solutions of Schroedinger's equation in the regions I, II and III can be expressed as

$\psi_I = A e^{\kappa x}$;

$\psi_{II} = +B\cos(kx)$ for even, and $-B\sin(kx)$ for odd;

$\psi_{III} = +Ae^{-\kappa x}$ for even, and $-Ae^{-\kappa x}$ for odd;

where $\kappa = \sqrt{\frac{-2mE}{\hbar^2}}$, and $k = \sqrt{\frac{2m(E+V_o)}{\hbar^2}}$.

The boundary conditions:

$\psi_I(-\frac{a}{2}) = \psi_{II}(-\frac{a}{2})$; $\psi_{II}(+\frac{a}{2}) = \psi_{III}(+\frac{a}{2})$; $\psi'_I(-\frac{a}{2}) = \psi'_{II}(-\frac{a}{2})$; $\psi'_{II}(+\frac{a}{2}) = \psi'_{III}(+\frac{a}{2})$;

will yield two transcendental equations,

$\tan(\frac{ka}{2}) = +\frac{\kappa}{k}$ (even) and $\cot(\frac{ka}{2}) = -\frac{\kappa}{k}$ (odd), or equivalently, $\tan\theta = \sqrt{\frac{\theta_o^2}{\theta^2} - 1}$ and $\cot\theta = -\sqrt{\frac{\theta_o^2}{\theta^2} - 1}$, where $\theta = \frac{ka}{2} = \sqrt{\frac{2m}{\hbar^2}(\mathrm{E} + \mathrm{V_o})}\,\frac{a}{2}$, $\theta_o = \sqrt{\frac{2m\mathrm{V_o}}{\hbar^2}}\,\frac{a}{2}$, and $\sqrt{\frac{\mathrm{V_o}}{\mathrm{E} + \mathrm{V_o}}} = \frac{\theta_o}{\theta}$.

These equations can only be solved numerically or graphically to obtain eigenvalues. The theory of finite square well potential has been applied to the ***Quantum Well*** or multilayer systems.

Conditions:

1. Let $f(\theta) = \tan(\theta) = \sqrt{\frac{\theta_o^2}{\theta^2} - 1}$.

For $f(\theta) = \tan(\theta)$: $\begin{cases} as \quad \theta \to 0, \quad f(\theta) \to 0 \\ as \quad \theta \to \frac{\pi}{2}, \quad f(\theta) \to \infty \end{cases}$, and

For $f(\theta) = \sqrt{\frac{\theta_o^2}{\theta^2} - 1}$: $\begin{cases} as \quad \theta \to 0, \quad f(\theta) \to \infty \\ as \quad \theta \to \theta_o, \quad f(\theta) \to 0 \end{cases}$.

This condition confirms that the two functions must intersect no matter how small θ_o is. Hence, there is always at least one bound state in a potential well.

2. The even and odd solution alternate as θ increases and there is one solution for each θ interval of $\frac{\pi}{2}$.

3. The solutions changes as V_o varies: at least one solution or one bound state in the 'even parity' graph even for a very small V_o.

4. As $V_o \to \infty$ the solution tend to multiples of $\frac{\pi}{2}$, i.e. $\tan(\frac{ka}{2}) = \infty$. This implies the solution of infinite square well potential, i.e. $\frac{ka}{2} = \frac{n\pi}{2}$ or $E = \frac{\pi^2 \hbar^2}{2ma^2} n^2$.

5. As a $\to$ 0 (i.e. well shrinks), and $V_o \to \infty$ to hold the constant ratio $a^2 V_o$ in $\theta_o = \sqrt{\frac{ma^2 V_o}{2\hbar^2}}$ or $\theta_o \propto a\sqrt{V_o}$. This gives the condition of a delta function potential.

The theory of finite square well potential can be applied in quantum well.

Appendix D

Mathieu potential

One-dimensional Mathieu problem yields many interesting features for describing the electronic structure of a condensed matter system. The problem is set up considering a periodic potential of the form $V(x) = 2V_0 \{(1 - \cos(2\pi x/L)\}$ with period L. The Schroedinger's equation is now called Mathieu's equation:

$$\frac{-\hbar^2}{2m}\frac{d^2u_x(x)}{dx^2} + V(x)\, u_x(x) = E_x u_x(x).$$

Using dimensionless variables such as

$$w = \pi x/L;\; s = 8mL^2 V_0/(\pi\hbar^2);\; \epsilon = \left(\frac{m}{2V_o}\right)^{1/2}\frac{LE}{\pi\hbar} \text{ then}$$

the Mathieu equation reduces to

$$d^2u/dw^2 + (\epsilon s^{1/2} - s/2 + \frac{1}{2}s\cos(2w))\,u = 0.$$

Expansion of V(x) about a minima, x=0, gives $V(x) \approx \frac{4\pi^2 V_o}{L^2}x^2$ with natural frequency

$\nu_o = \sqrt{\frac{2V_o}{mL^2}}$. This connects the Mathieu's equation with the free electron case of energy bands in a crystal. The potential is now a linear harmonic oscillator. The wavefunctions and the corresponding energy of such system would be

$$u_n = \exp(-z^2/2)\, H_n(z), \text{ and } E_n = (2n+1)\frac{1}{2}h\nu_o = \epsilon_n \frac{1}{2}h\nu_o \text{ respectively.}$$

Where $H_n(z) = (-1)^n \exp(z^2)\frac{d^n}{dz^n}\exp(-z^2/2)$, and $z = S^{\frac{1}{4}} w = S^{\frac{1}{4}}\pi x/L$.

We can conceive that $mv_o^2L^2\{(1 - \cos(2\pi x/L)\}$ (sinusoidal potential) reduces to $2\pi^2 mv_o^2x^2$ (quadratic potential) located at x = 0, ±L, ±2L etc; however*, it joins smoothly from one to the next space points*, with potential maximum in between as shown in Figure D.1.

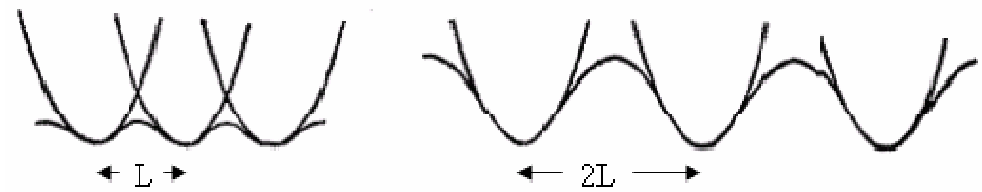


Figure D.1 Parabolic and sinusoidal potentials

The height of the maximum potential increases as L increases provided v_o constant.

Also, choosing z = 1;

$$x = x_o, \text{ and } S^{\frac{1}{4}} = 1/ w_o = L/ \pi x_o.$$

This shows that the dimensionless measure of lattice constant or L can be obtained from $S^{\frac{1}{4}}$. The Figure as given below shows the widths of energy bands and energy gaps of the one-dimenstional problem as a function of lattice spacing.

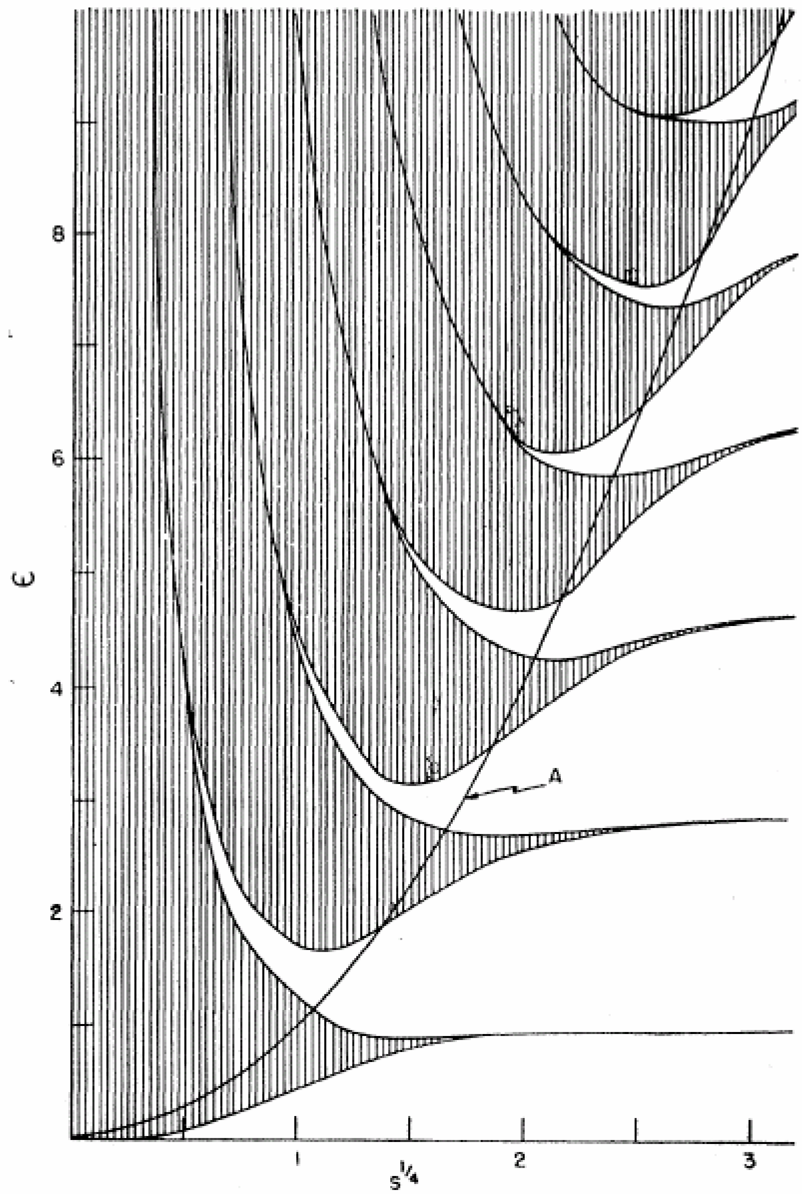


Figure D.2 Widths of energy bands and energy gaps of the one-dimensional problem as a function of lattice spacing. Curve A represents height of the potential barrier between atoms

Appendix E

Physics of the minibands in a superlattice

The Bloch oscillation is a central issue in the minibands of the semiconductor superlattice. We first discuss the case in the absence of scattering (an ideal situation) and then introduce scattering for a band electron. The semiclassical motion of the electron in the absence of scattering is described by the equations

$$\boxed{k(t) = -\frac{e}{\hbar}\varepsilon t, \quad \text{and} \quad v(t) = \frac{1}{\hbar}\left[\frac{dE(k)}{dk}\right]_{k=k(t)}}$$

The implications of the above relations in the free-electron case and for an electron in an actual energy band are of particular interest. In the free-electron case, the momentum, the velocity, and the kinetic energy of the electron increase indefinitely in time.

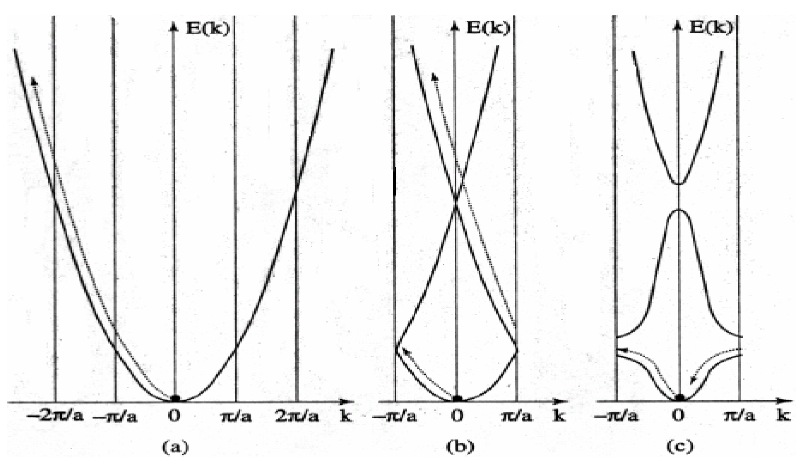


Figure E Motion of an electron in k-space under constant field ε in x-direction (a) free-electron case (b) folding case (a) into first Brillouin zone (c) fast oscillation of electron in the presence of the electric field and of a periodic potential [20]

The situation is radically different in an actual energy band, whenever interband transitions can be neglected. In the presence of an electric field and of a periodic potential, the electron velocity oscillates and the electron undergoes fast oscillating motion as shown in Figure E(c). For example, when an electric field acts on an electron at k = 0 (positive effective mass at this point), it first acquires energy and velocity. At the top of the band at k = $-\frac{\pi}{a}$ (at this point m* flips to -∞ as $\frac{d^2E(k)}{dk^2} \rightarrow 0$) the velocity ceases; as a result, there is no interband tunneling, but electron continues its trajectory on the same band from k = $+\frac{\pi}{a}$ (m* starts decreasing from this point) and it begins to lose energy until it reaches the initial state at k = 0. As the electron's k continues to change at a uniform rate in time, it moves up and down the band E(k). Such an ideal oscillatory motion of an electron in an interband of superlattice is called 'Bloch oscillations' and the electron is called 'Bloch oscillator' [20]. The Bloch frequency is $\omega_B = \frac{2\pi}{T_B} = \frac{e\mathcal{E}a}{\hbar}$.

$$\because \ \frac{2\pi}{a} = \frac{e}{\hbar}\, \varepsilon\, T_{B.}$$

The Wannier-Stark ladder is another important phenomenon that occurs in superlattice energy spectrum.

Appendix F

Dimensionality and Quantum Confinement

The confinement effect gives rise to low dimensionality in quantum well(QW), quantum well wire (QWW) and quantum dot (QD) because the propagation of carriers is restricted to less than 3 dimensions in such structures. This will dramatically modify the electron energy spectrum or density of states (DOS) near the band edge.

If we assume an infinite confining potential, i.e. $V(z) = \begin{cases} 0 & |z| \leq \frac{L}{2} \\ \infty & \text{elsewhere} \end{cases}$, then the allowed energy states and the corresponding wavefunctions can be presented as [31]:

QW: $$E_n(k_x, k_y) = \frac{\pi^2 \hbar^2 n^2}{2m^* L_z^2} + \frac{\hbar^2}{2m^*}(k_x^2 + k_y^2),\ \psi = \phi(z)\exp(ik_x x + ik_y y);$$

QWW: $$E_{n,m}(k_x) = \frac{\pi^2 \hbar^2}{2m^*}\left(\frac{n^2}{L_z^2} + \frac{m^2}{L_y^2}\right) + \frac{\hbar^2 k_x^2}{2m^*},\ \psi = \phi(z)\phi(y)\exp(ik_x x);$$

QD: $$E_{n,m,l} = \frac{\pi^2 \hbar^2}{2m^*}\left(\frac{n^2}{L_z^2} + \frac{m^2}{L_y^2} + \frac{l^2}{L_x^2}\right),\ \psi = \phi(z)\phi(y)\phi(x).$$

where n, m, l = 1, 2, … are the quantum confinement numbers; L_z, L_y and L_x are the confining dimensions; we can approximate

$\phi(z) = \sqrt{\frac{2}{L_z}}\sin(\frac{\pi n}{L_z} z)$, $\phi(y) = \sqrt{\frac{2}{L_y}}\sin(\frac{\pi n}{L_y} y)$ and $\phi(x) = \sqrt{\frac{2}{L_x}}\sin(\frac{\pi n}{L_x} x)$. The term $\exp(ik_x x + ik_y y)$ is the wavefunction describing the electronic motion in directions x and y, similar to free electrons wavefunctions.

Appendix G

Dynamics of an electron in band theory

Quasi-momentum

In the free electron model (constant potential case), the plane wave $W(k, x) = (1/\sqrt{L}) \exp(ikx)$ and the energy $E(k) = \hbar^2k^2/2m$ are the eigenfunctions and eigenvalues respectively. The momentum operator $p = -i \hbar\, d/dx$ represents the momentum of an electron, because $pW(k, x) = \hbar k\, W(k, x)$, i.e. the plane wave is the eigenfunction of p with eigenvalue $\hbar k$.

In a periodic potential, $\vec{k}$ varies in Brillouin zone and the eigenfuction is indicated by Bloch function as $\psi(k, x) = \exp(ikx)\, u(k, x)$. However, $\psi(k, x)$ is not eigenfunction of p, because $-i \hbar\, d/dx\, (\psi(k, x)) = -i \hbar\, d/dx[\exp(ikx)\, u(k, x)] = \hbar k\, \psi(k, x) + \exp(ikx)\, (-i)\, \hbar\, d/dx\, (u(k, x)) \neq \hbar k\, \psi(k,x)$.

Hence, $\hbar k$ is **not the momentum of the electron** in the state $\psi(k,x)$**, but crystal momentum** . This also implies that the eigenfunctions of the crystal Hamiltonian are not eigenfunctions of the p operator (except in the empty lattice situation). In other words, in general, a Bloch wavefunction $\psi(k,x)$ is not an eigenfunction of the momentum operator $p = -i \hbar\, d/dx$. However, $\psi(k,x)$ can be expressed as a linear combination of plane waves of wavenumbers k, $k \pm 2\pi/a$, , $k \pm 4\pi/a$ etc. Thus, the only possible values of a measure of the observable p on the Bloch function $\psi(k,x)$ are $\hbar k$, $\hbar k \pm \hbar (2\pi/a)$, , $\hbar k \pm \hbar (4\pi/a)$ etc. For this reason, the quantity $\hbar k$ is called **quasi-momentum** of the electron in the crystal, or also **crystal momentum** [20].

Velocity

The semiclassical electron velocity in the state ψ(k,x) is defined as

$$v(k) = \left\langle \psi(k, x) \,|\, \frac{p}{m} \,|\psi(k, x) \right\rangle = \left\langle u(k, x) \,|\, \frac{1}{m}(p + \hbar k) \,|u(k, x) \right\rangle ,$$

and likewise the dispersion relation from Schroedinger's equation is

$$E(k) = \left\langle \psi(k, x) \,|\, \frac{p^2}{2m} + V(x) \,|\psi(k, x) \right\rangle ,$$

where $\psi(k, x) = \exp(ik\, x)\, u(k, x)$ is the crystal wavefunction in the Bloch form. With manipulation and normalization of u* and u to 1, we get,

$$\frac{dE(k)}{dk} = \left\langle u(k, x) \,|\, \frac{\hbar}{m}(p + \hbar k) \,|u(k, x) \right\rangle .$$

Eventually, we get,

$v(k) = \dfrac{1}{\hbar} \dfrac{dE(k)}{dk}$. This is, in fact, called the group velocity of the electron.

Effective mass

When an external electric field $\vec{E}$ is imposed on a free electron, it experiences a force $\vec{F} = -e\vec{E}$. So, the acceleration of an electron is $\vec{a} = \dfrac{-e\vec{E}}{m}$. Surprisingly, the electron accelerates more rapidly in the solid than in free space. As we know, classically, E(k) = T(k) + U(k); where the potential U(k) is constant in <u>free space</u> and hence only the kinetic energy varies with wavevector $\vec{k}$. However, in a <u>solid,</u> due to the periodic field of the lattice, sum of the both k-dependent kinetic and k-dependent potential energy contribute to the total energy of any state. This is the origin of 'effective mass' definition of an electron in a more general non-parabolic band dispersion of solid.

Since the electron is represented by a wavepacket (a superposition of a set of plane waves i.e. electron wavefunction), its velocity is defined as the group velocity [10],

$$v_g(k) = \frac{d\omega(k)}{dk} = \frac{1}{\hbar}\frac{dE(k)}{dk}; \qquad \frac{dv_g(k)}{dt} = \frac{1}{\hbar}\frac{d^2E(k)}{dkdt} = \frac{1}{\hbar}\frac{d^2E(k)}{dk^2}\frac{dk}{dt}.$$

On the other hand, the work done by the external force to a band electron is,

$$\delta E(k) = F\, v_g(k)\, \delta t \equiv \frac{dE(k)}{dk}\delta k = \hbar\, v_g(k)\, \delta k.$$

This further implies,

$$F = \hbar\frac{dk}{dt} \equiv m^* \frac{dv_g(k)}{dt} = m^* \frac{1}{\hbar}\frac{d^2E(k)}{dk^2}\frac{dk}{dt} \text{ [10].}$$

To describe the motion of band electrons subjected to an external force, the effective mass is substituted for the inertial mass of the relation “force = mass x acceleration.”

Therefore, $m^* = \dfrac{\hbar^2}{\left(\dfrac{d^2E(k)}{dk^2}\right)}$, the effective mass of band electron.

Here, the effective mass is energy dependent as given by the energy dispersion $E_j(\vec{k})$; particularly, it is useful to deal in the regions with almost empty or almost full bands, i.e. the states close to the minima and maxima. These regions are approximately parabolic. The equation relates the effective mass of an electron directly to the local curvature of the energy band at the specified $\vec{k}$. This will allow us to deal with carriers (electrons or holes) whose effective mass is much different from the bare electron mass, and also with negative effective mass [20].

In 3D:

$$a_i = \sum_j \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_i \partial k_j} F_j, \qquad i, j = x,y,z$$

$$\left(\frac{1}{m^*}\right)_{ij} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_i \partial k_j}, \qquad \text{i , j = x,y,z}$$

$$E(k) = (\alpha_1 k_x^2 + \alpha_2 k_y^2 + \alpha_3 k_z^2), \qquad \text{i , j = x,y,z}$$

The effective mass is therefore a tensor.

Effect of external electric field

Consider the effect of an external electric field, ε (in positive x-direction), on the dynamics of an electron in a periodic potential V(x). The total Hamiltonian of the system is $H = \frac{p^2}{2m} + V(x) + e\,\varepsilon\,x$,

where e is absolute value of the electronic charge. With the time $t = t_o \equiv 0$, the wavenumber $k = k_o$; the time evolution of Bloch state $\psi(k_o, x)$ is given as

$$\psi(x, t; \varepsilon) = \exp(-\frac{i}{\hbar} Ht)\ \psi(k_o,x).$$

Replacing x by x + a in both expressions above, i.e. translation operation, and employing the periodic property of the functions such as

$$V(x + a) = V(x), \text{ and } \psi(k_o, x + a) = \exp(i\, k_o a)\ \psi(k_o,x),$$

then the time evolved Bloch type wavefunction will be constructed as

$$\psi(x + a, t; \varepsilon) = \exp(ik(t)a)\ \psi(x, t; \varepsilon);$$

the wavenumber, k(t), changes linearly in time t as given by

$k(t) = -\frac{e}{\hbar}\ \varepsilon\, t + k_o$.. Setting $k_o = 0$ at time t = 0, it yields

$$k(t) = -\frac{e}{\hbar}\ \varepsilon\, t.$$

REFERENCES

1. B. R. Nag, *Physics of Quantum Well Devices*, Kluwer Academic Publishers, 2000
2. J Shen, J P Pierce, E W Plummer and J Kirschner, *J. Phys.: Condens. Matter*, **15** (2003), pp. R13
3. J. M. D. Coey and S. Sanvito, *J. Phys. D: Appl. Phys.*, **37** (2004), pp. 988-993
4. M. Kondo, K. Maruyama and K. Kurihara, *FUJITSU Sci. Tech. J.,* **38** (2002), pp. 46-53
5. W. E. Pickett and J. S. Moodera, *Physics Today*, **54**(2001), pp. 39
6. J. Y. T. Wei, N. –C. Yeh, R. P. Vasquez and A. Gupta, *J. of Appl. Phys.,* **83** (1998), pp. 7366-7368
7. C. M. Fang, G. A. de Wijs and R. A. de Groot, *J. Appl. Phys.*, **91** (2002), pp. 8340-8344
8. P. Lugli, A. D. Carlo and A. Reale, *J. Phys.: Condens. Matter*, **11** (1999), pp. 6035-6043
9. M. Jaros, *Physics and Applications of Semiconductor Microstructures*, Oxford University Press, 1990
10. J. Singleton, *Band Theory and Electronic Properties of Solids*, Oxford University Press, 2001
11. Paul Harrison, *Quantum Wells, Wires and Dots*, John Wiley & Sons Ltd, 2002
12. H. F. Jones, Physics Department, Imperial College, London SW7 2BZ, UK (sp)

13. A. G. U. Perera and M. H. Francombe, Department of Physics and Astronomy, Georgia State Univ., Atlanta, Ga 30303 ; J, -W Choe, Department of Physics, Kyung Hee Univ., Suwon 449-701, Korea (WWW document)

14. J. I. Gersten and F. W. Smith, *The Physics and Chemistry of Materials*, John Wiley & Sons, Inc., 2001

15. H. M. Rosenberg, *The Solid State*, 3rd Ed., Oxford University Press, 2003

16. D. L. Aronstein and C. R. Stroud, *Phys. Rev. A*, **55** (1997), pp. 4526-4537

17. J. K. Boyd, *Journal of Mathematical Physics*, **42** (2001), pp. 15

18. B. I. Barker, G. H. Rayborn, J. W. loup and G. E. loup, *Am. J. Phys.,* **59** (1991) , pp. 1038-1042

19. Federico Capasso, *Physics of Quantum Electron Devices*, Springer-Verlag Berlin Heidelberg, 1990

20. G. Grosso and G. P. Parravicini, *Solid State Physics*, Academic Press, 2000

21. N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Holt, Rinehart and Winston, 1976

22. T. N. Todorov, *J. Phys. Cond. Matter*, **14** (2002), pp.3049

23. M. T. Dove, *Structure and Dynamics*, Oxford University Press, 2003

24. J. J. Sakurai, *Modern Quantum Mechanics*, Pearson Education Asia, 1994

25. P. M. Mathews and K Venkatesan, *A Textbook of Quantum Mechanics*, Tata McGraw-Hill Publishing Com. Ltd., New Delhi, 1990

26. M. Ali Omar, *Elementary Solid State Physics*, Addison-Wesley Publishing Comp., Inc., 1975

27. G. Burns, *Solid State Physics*, Academic Press, Inc., 1985

28. Peter S. Zory, *Quantum well lasers*, Academic Press Inc, 1993

29. C. Kittel, *Introduction to Solid State Physics*, 7th Ed., Wiley Eastern Limited, 1995

30. S. R. Elliott, *The Physics and Chemistry of Solids*, John Wiley & Sons, 1998

31. E. O'Reilly, *Quantum Theory of Solids*, Taylor & Francis, 2002

32. T. G. Dargam, R. B. Capaz and B. Koiller, *Phys. Rev. B*, **56** (1997), pp. 9625-9629

33. M. Altarelli, European Synchrotron Radiation Facility, B.P. 220, F-38042 Grenoble Cedex, France (seminar paper)

34. Michael de Podesta, *Understanding the properties of matter*, UCL Press, 1996

35. Jean-Marc Levy-Leblond and F. Balibar, *Quantics*, North-Holland, 1990

36. M. Fornari *et. al., Phys. Rev.,* **B55** (1997), pp.16339

37. C. H. Ahn, K. M. Rabe and J. –M. Triscone, *Science*, **303** (2004), pp. 488-491

38. E. P. Pokatilov *et. al.,* J. Phys.: Condens. Matter, **12** (2000), pp 9037-9052

39. V. Gasparian and M. Ortuno, *Physical Review B*, **56** (1997), pp. 14929-14932

40. Fuad Badrieh and D. K. Ferry, Dept. of Electrical Eng. & Center for Solid State Electronics Research, Arizona State University (seminar paper)

41. D. J. Singh and D. A. Papaconstantopoulos, *Electronic Structure and Magnetism of Complex Materials*, Springer Verlag, 2003

42. R. E. Cohen, *Physical Review B*, **50** (1994), pp. 14694-14697

43. H. J. Gotsis, D. A. Papaconstantopoulos, and M. J. Mehl, *Physical Review B*, **65** (2002), pp. 134101-1 to 134101-7

44. S. H. Yang *et. all.*, *Physical Review B*, **57** (1998), pp. R2013-R2016